Module 5

DC DRIVES

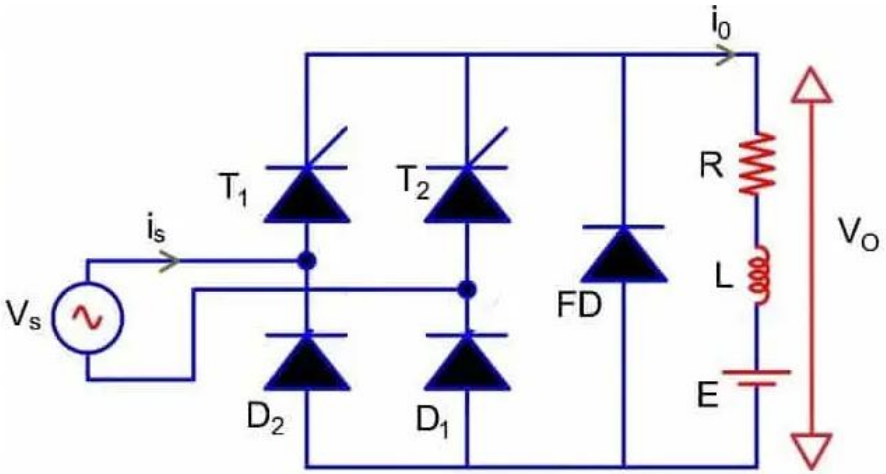
Single phase semi converter and single phase fully controlled converter drives.

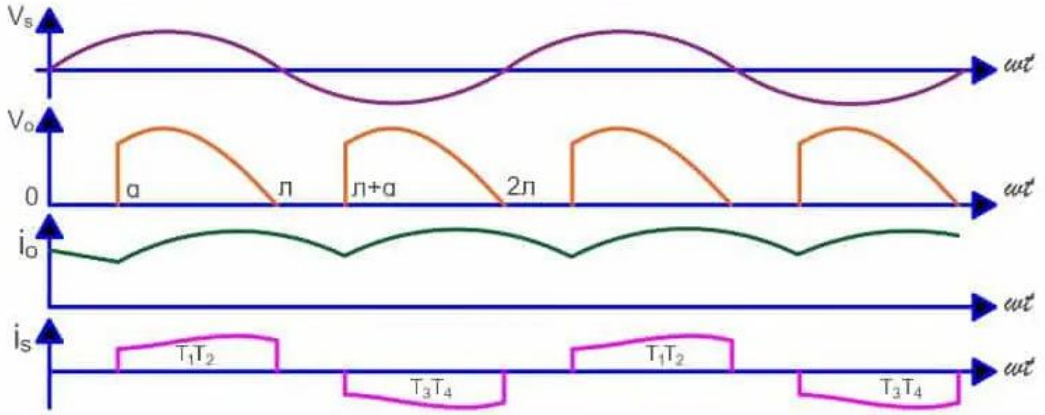
Dual Converters for Speed control of DC motor-1-phase and 3-phase configurations; Simultaneous and Non-simultaneous operation.

Chopper controlled DC drives- Single quadrant chopper drives- Regenerative braking control- Two quadrant chopper drives- Four quadrant chopper drives

SINGLE-PHASE SEMICONVERTER DRIVE

- Fig. shows a DC motor fed through a single phase semiconverters.
- This converter offers one-quadrant operation and used upto about 15kW.





SINGLE-PHASE SEMICONVERTER DRIVE

- The bridge circuit of the semi converter has two diodes (D1, D2) and two SCRs(T1, T2).
- Pair of a diode and SCR(D2, T1) is connected in the parallel with another pair of a diode and SCR(D1, T2).
- A free-wheeling diode (FD) has its connection in parallel with the load and bridge circuit.
- The freewheeling diode freewheels the energy stored in the inductor and, thus limits the overvoltage.

SINGLE-PHASE SEMICONVERTER DRIVE

From ωt=0 to ωt=π

- During the interval of ωt =0 to ωt=α, SCR T1 can be switched on by applying the firing pulse to its gate.
- During this period the SCR T1 and diode D1 come into conduction mode.
- After ωt=0, SCR T1 becomes forward biased provided $V_m \sin\omega t$ is greater than the back emf (E)of the motor.
- Let, SCR triggers at ωt=α. Under this condition, the SCR T1 conducts if $V_m \sin\alpha > E$, and the load current i_o flows through the RLE load through thyristor T1 and diode D1.
- The SCR T1 conducts from ωt =α to ωt=π. The thyristor T1 switches off during ωt =π to ωt=2π because of the negative voltage at its anode.
- The output voltage at the load terminal during ωt=α to ωt=π is just a replica of the source voltage(V_S).

SINGLE-PHASE SEMICONVERTER DRIVE

Just after ωt =π to ωt=2π

- The supply voltage reverses and becomes negative just after ωt=π. The anode of SCR T_1 receives a negative supply at its anode and it goes into the off state.
- The reversal of supply voltage causes forward biasing of a free-wheeling diode(FD). In this condition, the load current io flow from T1D1 to FD.
- The output voltage across the load becomes zero at ωt=π because of the conduction of the freewheeling diode.
- During ωt =π to ωt=2π, the SCR T2 is in reverse bias if the source voltage is more than the E.
- For ωt=π+α, source voltage (Vs) is greater than E, T2 turns on, and the source current flow from FD to T2 D2. FD gets reverse biased just after turning on SCR T2.
- Just after ωt=2π, the SCR T_2 turns off and at the same time, the freewheeling diode FD gets forward biased. Now, the load current i_o gets transferred to FD.

SINGLE-PHASE SEMICONVERTER DRIVE

$$\text{for a 1- phase semiconverter, average output voltage, } V_0 = V_t = \frac{V_m}{\pi}(1+\cos\alpha)$$

α is the firing angle for semiconverter in the armature circuit

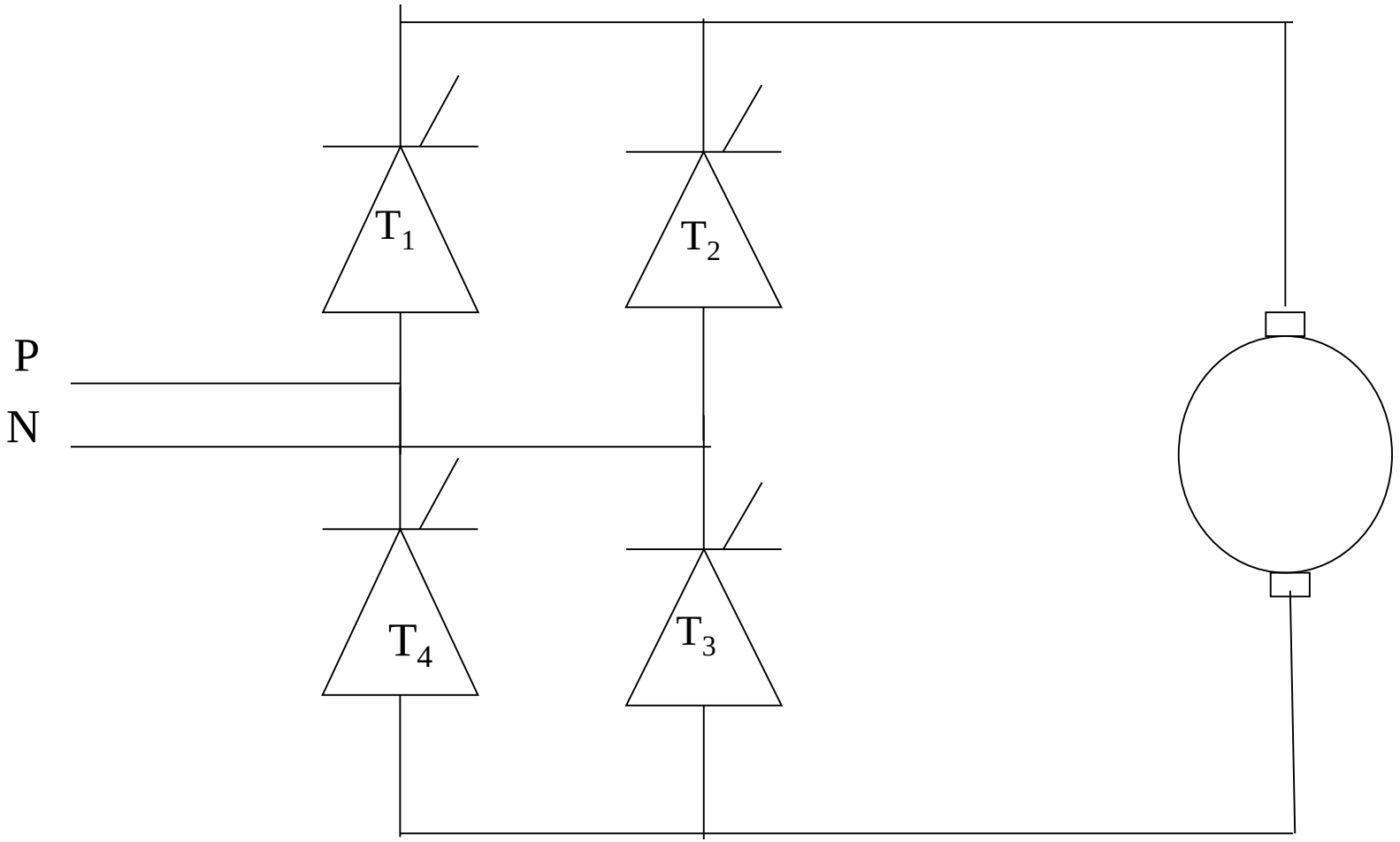
$$\text{RMS value of source current} = I_{sr} = I_a\left[\frac{\pi - \alpha}{\pi}\right]^{1/2}$$

$$\text{RMS value of thyristor current} = I_{Tr} = I_a\left[\frac{\pi - \alpha}{2\pi}\right]^{1/2}$$

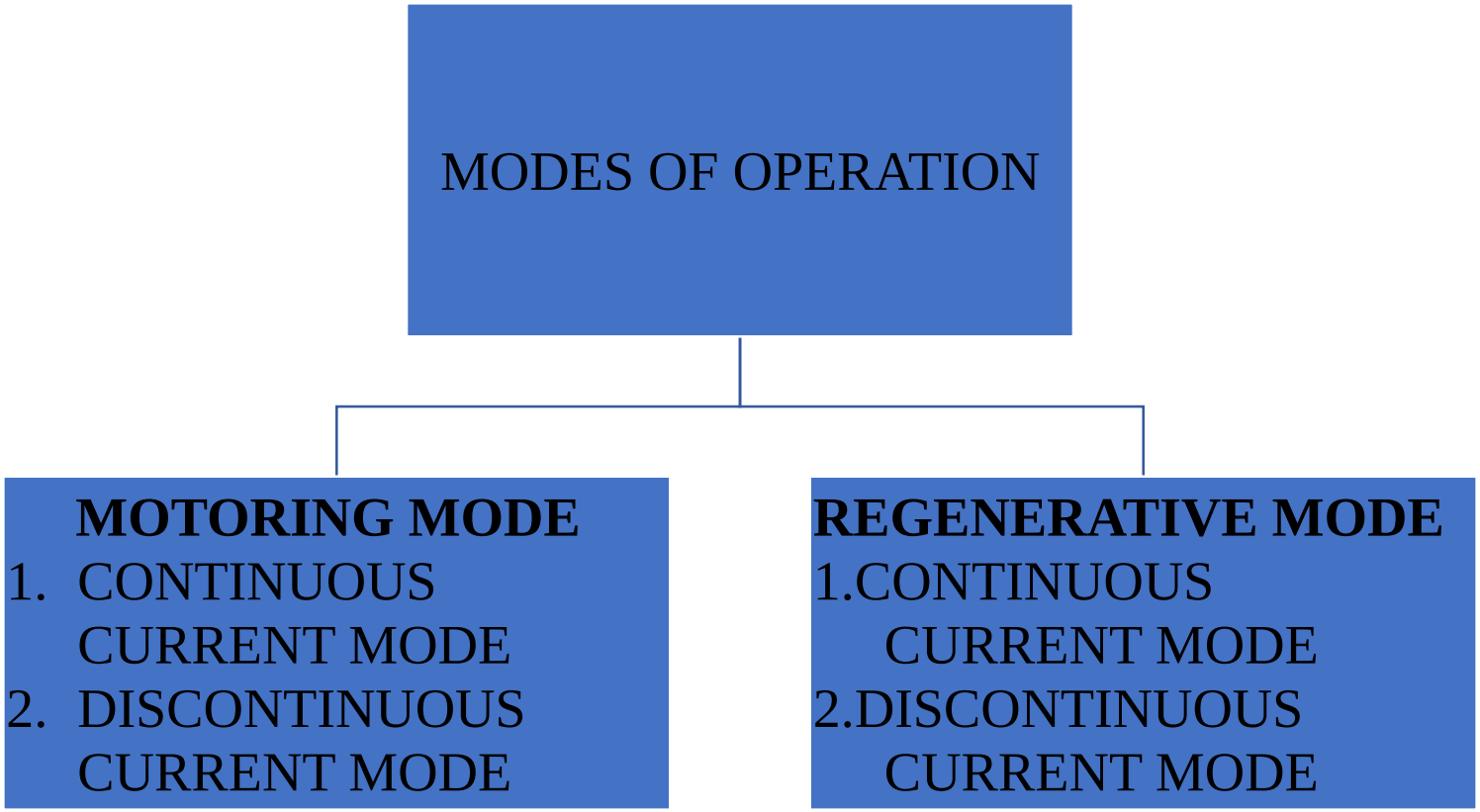
$$\text{Input pf} = \frac{V_t I_a}{V_s I_{sr}}$$

SINGLE PHASE FULLY CONTROLLED CONVERTER FED DC MODRIVES

Fully Controlled Single Phase rectifier



MODES OF OPERATION

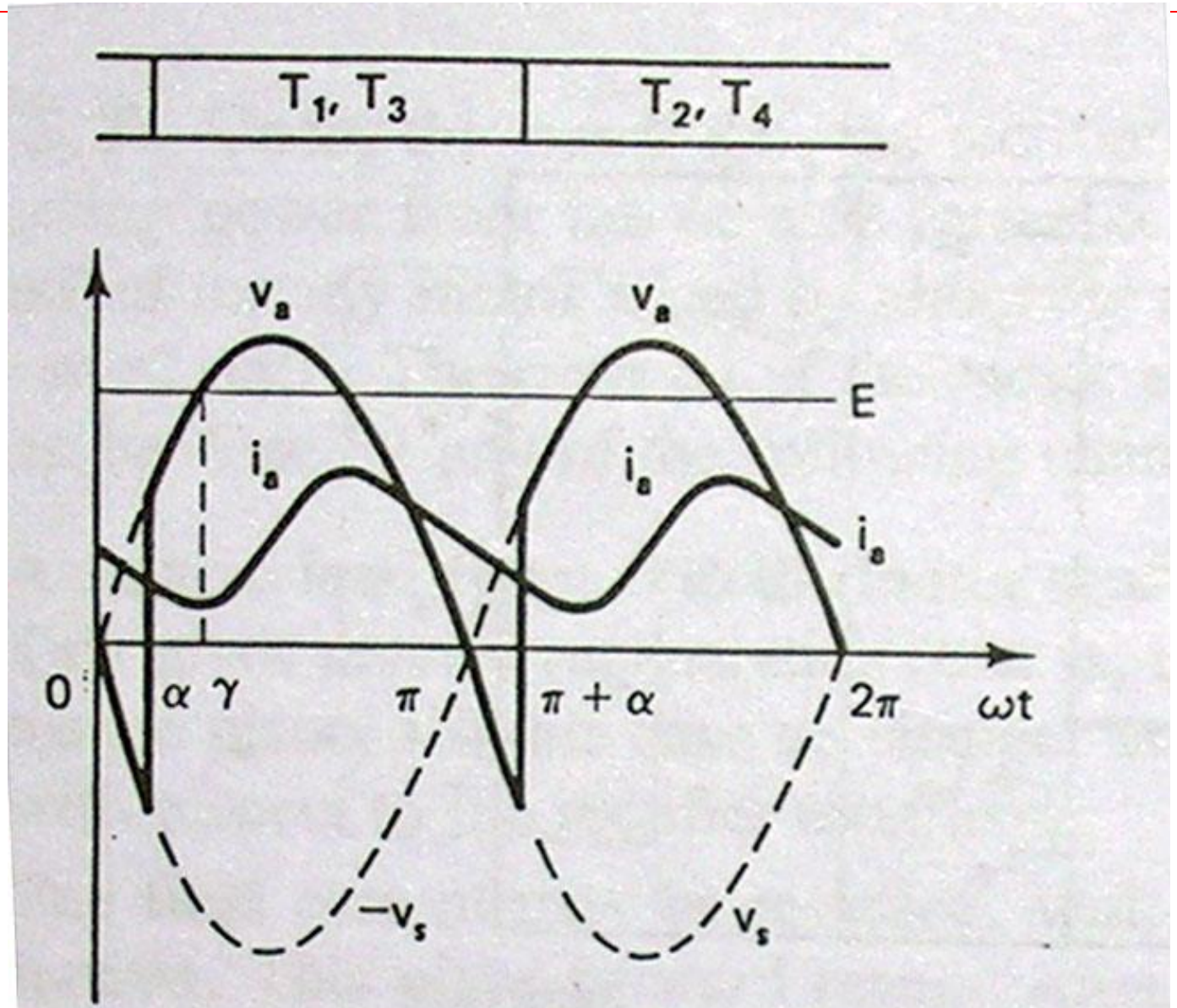


Modes of Operation

Notations & Symbols.

V_m	The peak Value of Supply Voltage.
ω	Supply Frequency
β	the angle atr which armature current equal to zero(extinction Angle)
$\beta 1$	β-л.
γ	$\sin^{-1}(E/V_m)$,angle at which source voltage equals to back emf.
γ'	л- sin-1(E/Vm)

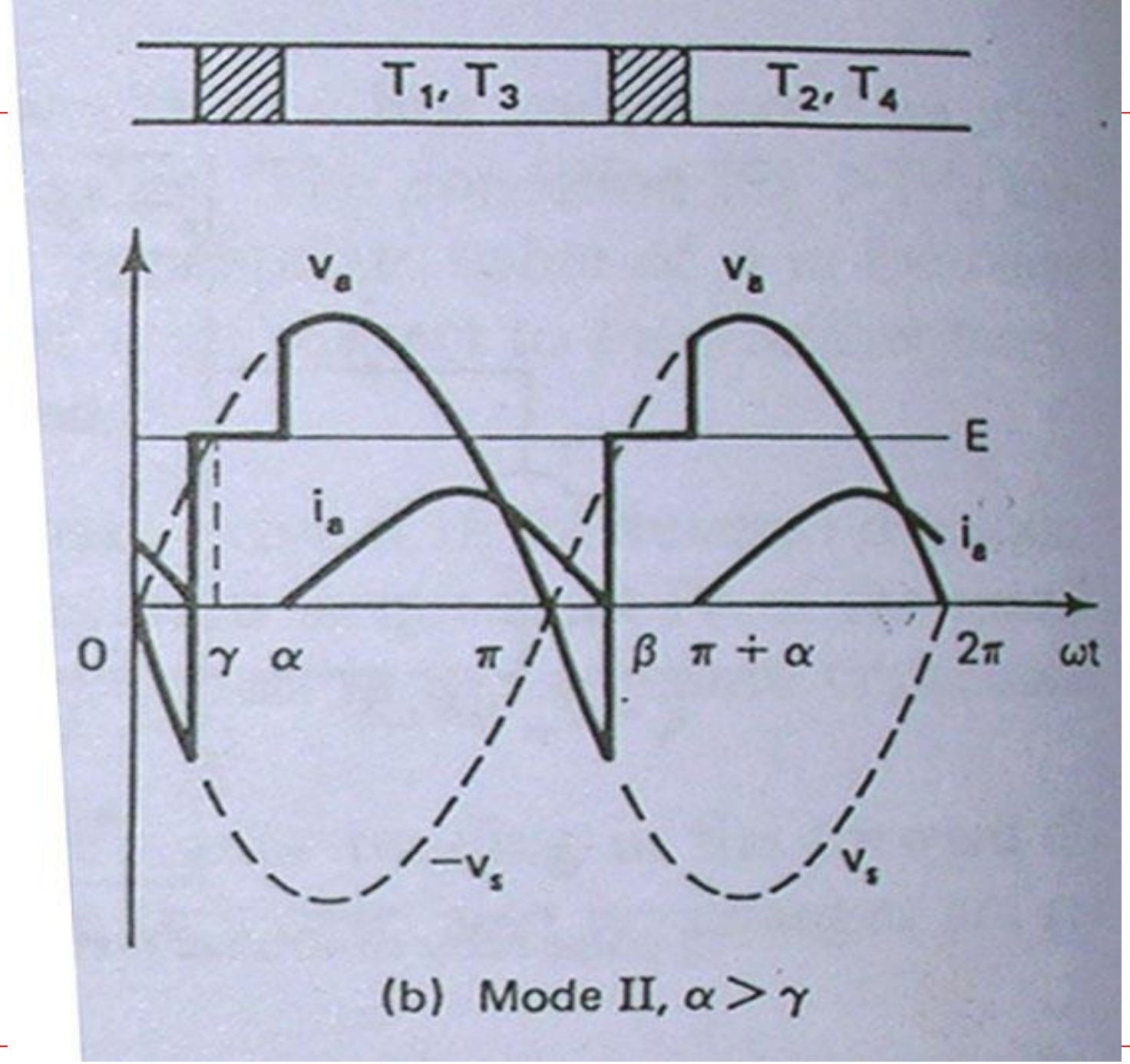
- The motor armature current flows through either T_1,T_3 or T_2,T_4
- When T1,T3 conduct voltage equals to Vs and when T2,T4 conducts Va= -Vs.
- If Ia =0; Va =E.
- When Ia >0,at the instant of triggering the thyristor ,then the biasing on it is decided by the source voltage only. i.e. the thyristor will start conducting even if E.>Vs.
- When Ia=0,at the instant of triggering the thyristor ,then the biasing on it is decided by the difference of source voltage and the back emf.. i.e. the thyristor will start conducting Only whenit Vs>E.



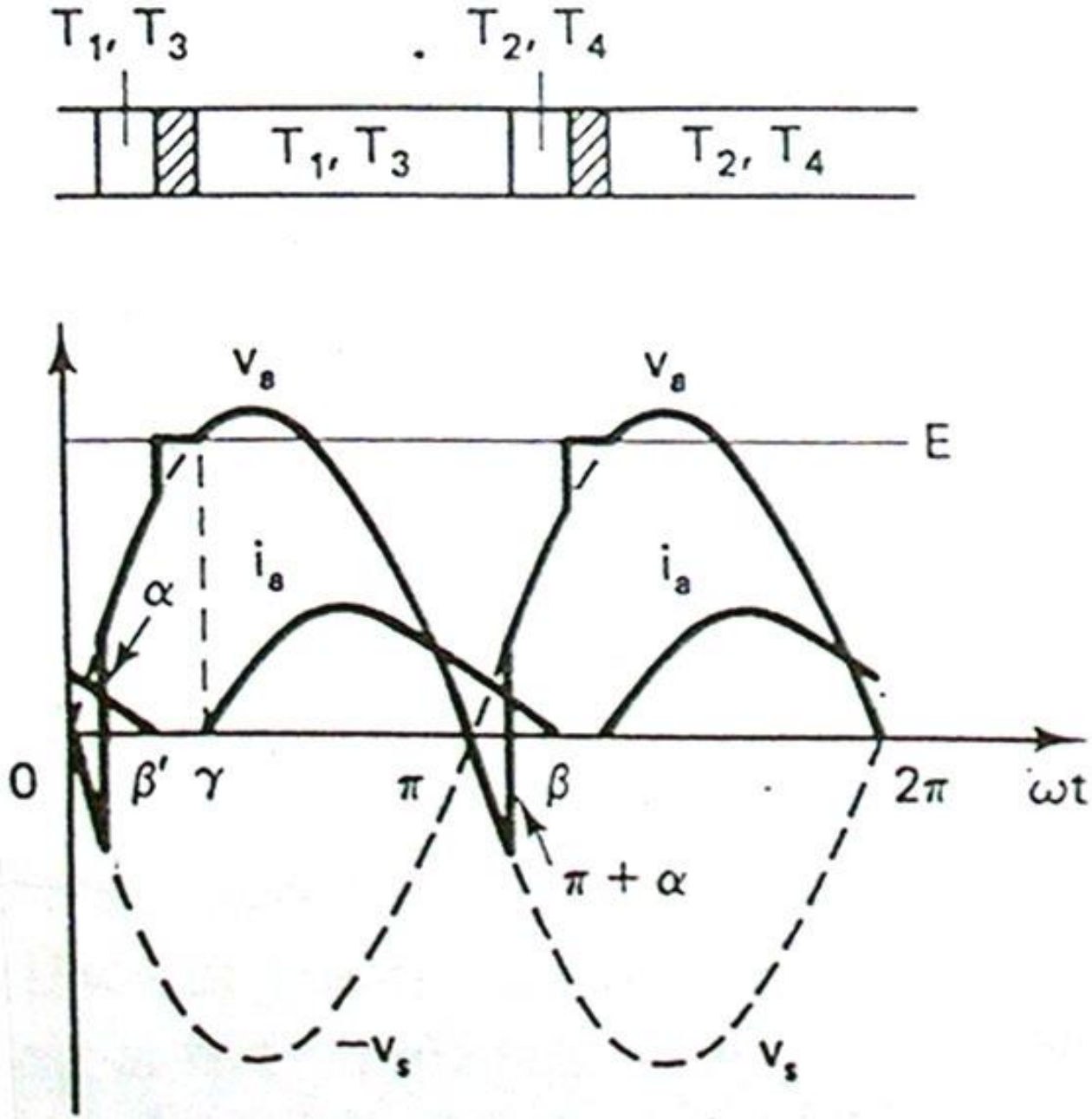
(a) Mode I, $\alpha >$ or $< \gamma$

MODE-I

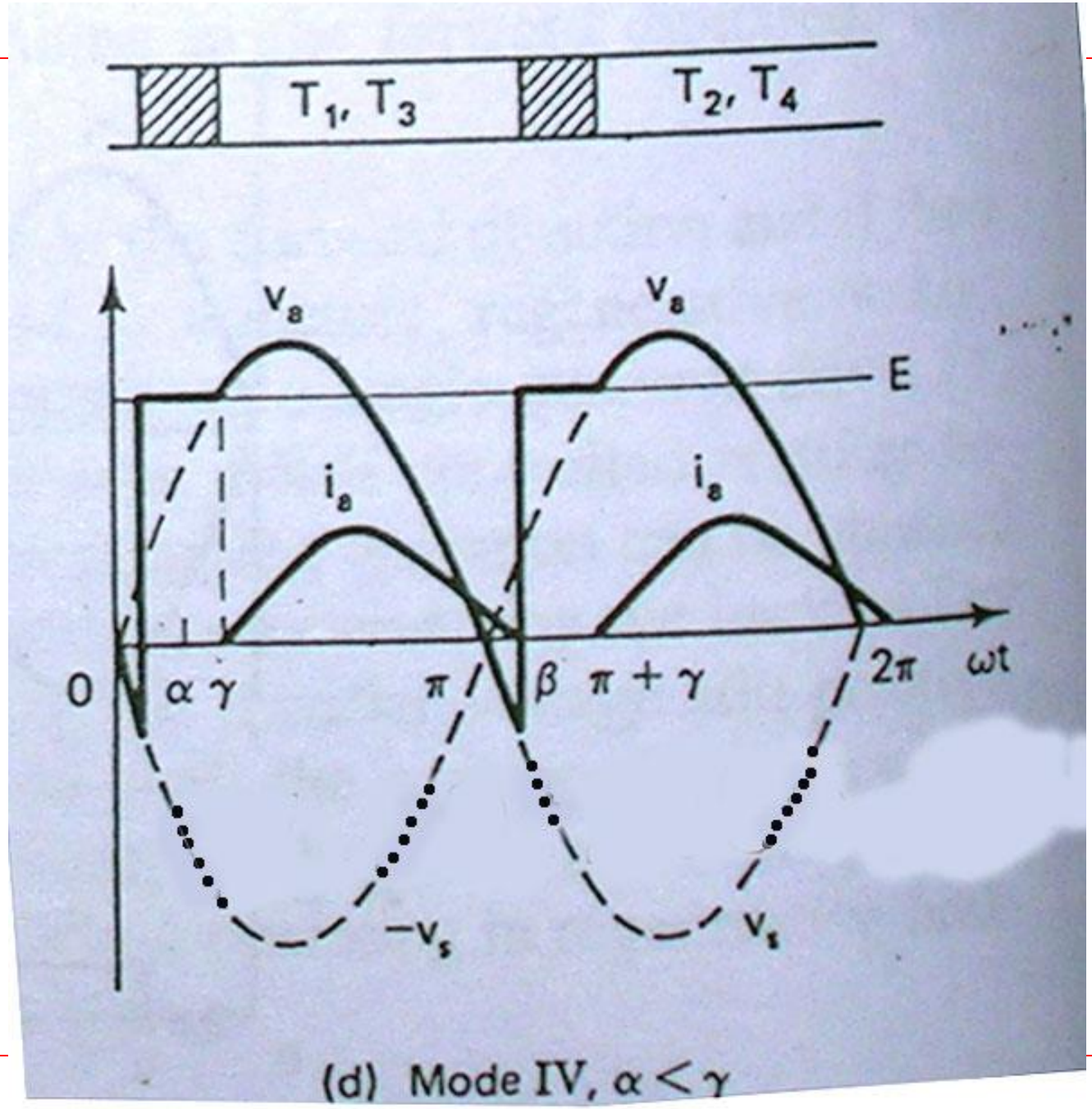
- ✓This is a continuous current mode because the armature current is continuous
- ✓The firing angle (α)may be greater than or less than γ
- ✓The drives will be in continuous current operation only for large values of inductance ,small values of firing angle and large value of current.



(b) Mode II, $\alpha > \gamma$



(c) Mode III, $\alpha < \gamma$



(d) Mode IV, $\alpha < \gamma$

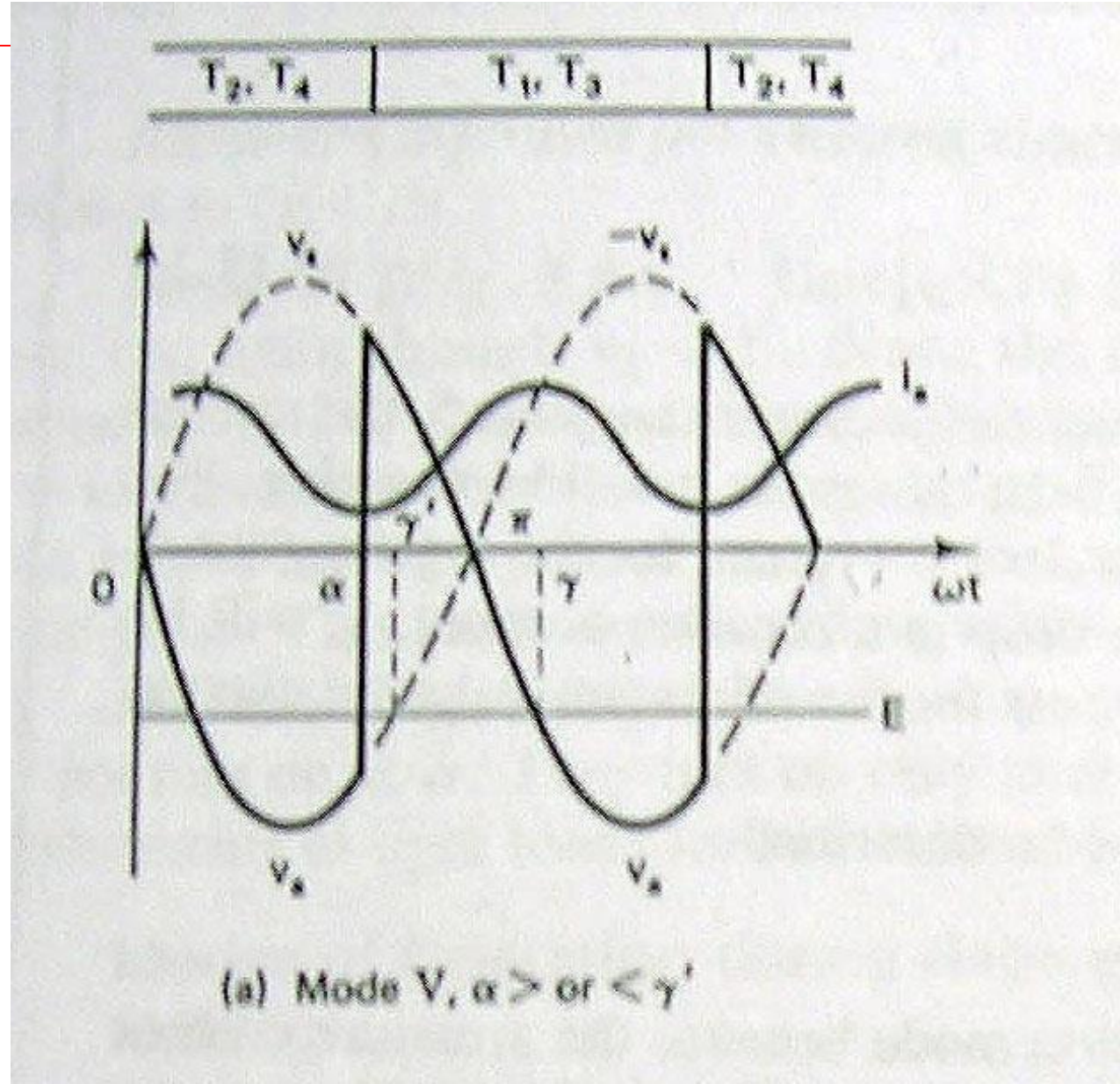
Mode I to IV are Motoring Modes.

Mode I is continuos current mode, $\alpha >$ or $< \gamma$

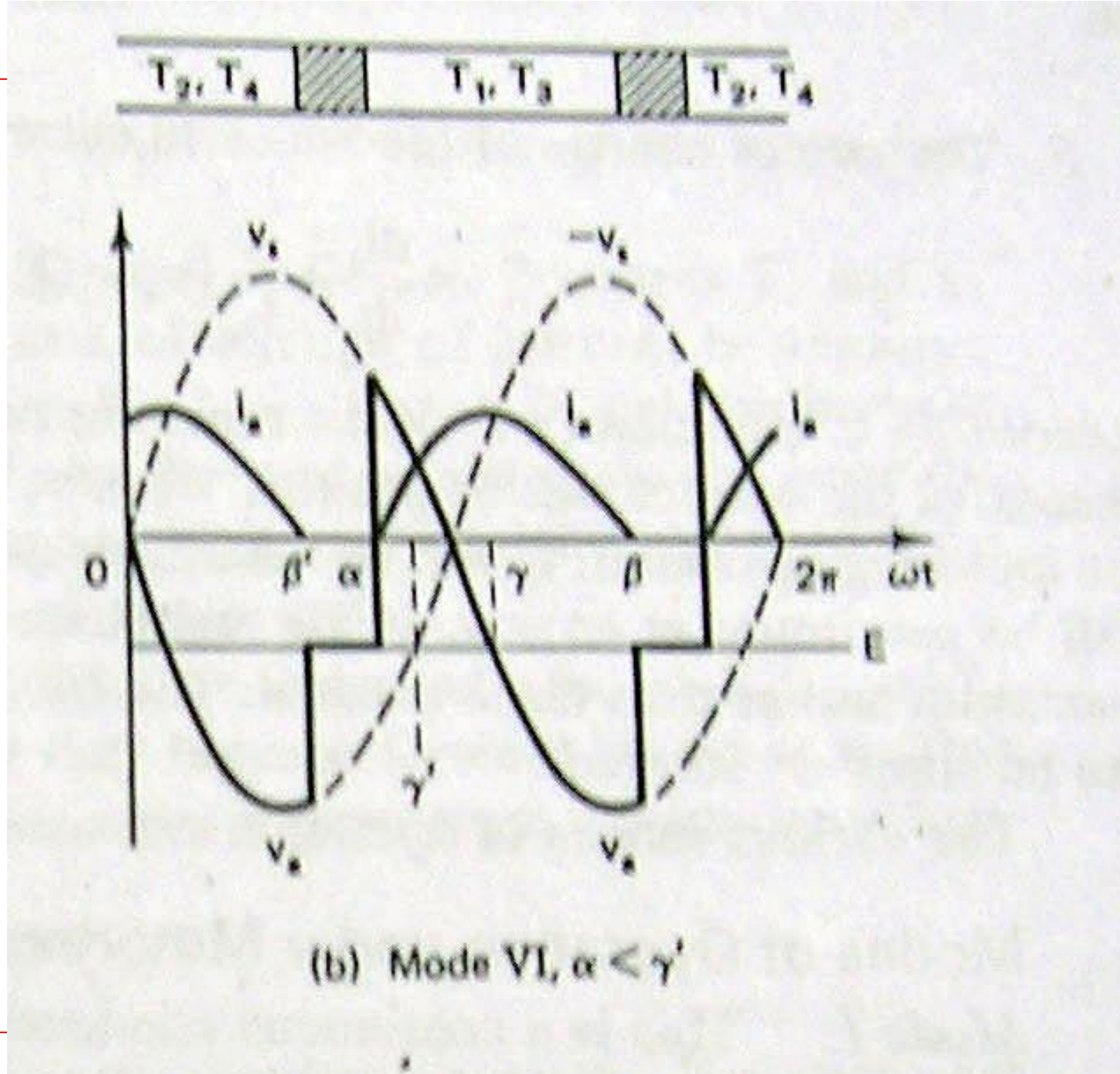
Mode II discontinuous mode of operation $\alpha > \gamma$

Mode III discontinuous mode of operation $\alpha < \gamma, \beta$

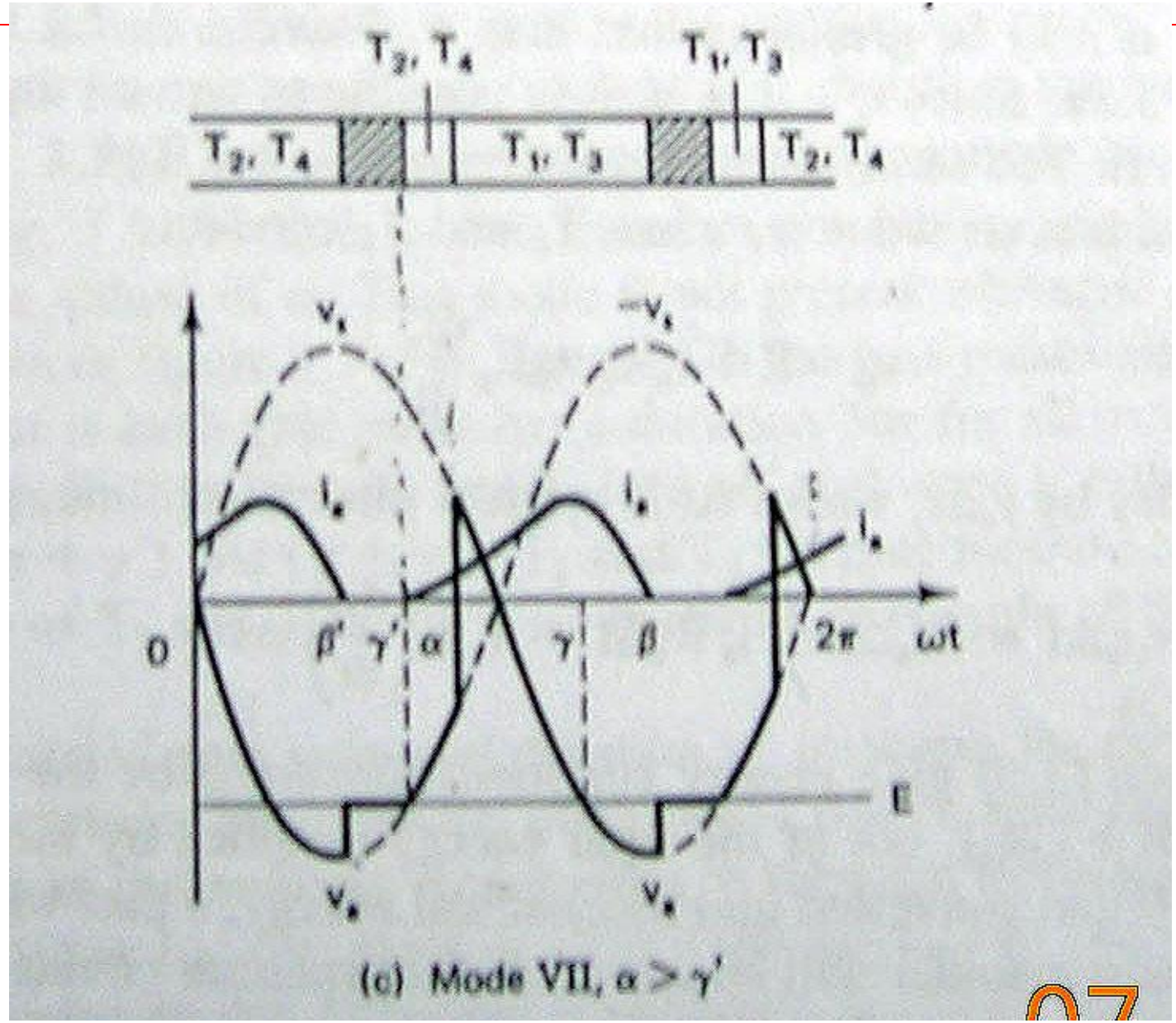
Mode IV discontinuous mode of operation $\alpha < \gamma$, $\alpha > \beta$



(a) Mode V, $\alpha >$ or $< \gamma'$



(b) Mode VI, $\alpha < \gamma'$



(c) Mode VII, $\alpha > \gamma'$

Mode V,VI,VII are regenerative operation mode

Mode V is a continuous current operation mode.

It occurs for large torques.

for low power drives that occurs for torque greater

than rated torques.

Mode VI is a discontinuous mode $\alpha < \gamma$

Mode VII is also a discontinuous mode.

NOTE:-REMEMBER THE WAVEFORMS OF AT LEAST TWO MODES FROM MOTORING AND REGENERATIVE MODES OF OPERATIONS

ANALYSIS

Duty Interval

When any one pair of thyristor conducts, AC source is connected to the motor. In this case motor terminal voltage is V_s when T_1, T_3 are in the conduction mode. For T_2, T_4 it is $-V_s$

$$V_a = L_a \frac{di}{dt} + R_a i_a + K\omega_m = V_m \sin \omega t, \ldots\ldots when\, T_1, T_3 \, are ON$$

$$V_a = L_a \frac{di}{dt} + R_a i_a + K\omega_m = -V_m \sin \omega t \ldots\ldots when\, T_2, T_4 \, are ON$$

Zero Current Interval.

Ia=0.

Va=E.=$K\omega_m$

MODE-I

Current through drive consists of

- component due to ac source
- component due to back emf
- transient component

$$i_a(\omega t) = \frac{V_m}{Z}\sin(\omega t - \psi) - \frac{K\omega_m}{R_a} + k_1 \exp(\frac{-t}{\tau_a})$$

$$Z = [R_a{}^2 + (\omega La)^2]^{1/2}$$

$$\tau_a = \frac{L_a}{R_a}$$

$$\psi = \tan^{-1}\frac{\omega L_a}{R_a}$$

Now find K_1,

Use the condition $i_{a(}(\alpha)=i_a(л+\alpha)$

$$i_a(\omega t) = \frac{Vm}{Z}\left[\sin(\omega t - \psi) - \frac{2\sin(\alpha-\psi)\exp\{(\alpha-\omega t)\cot\psi\}}{1-\exp(-\pi\cot\psi)}\right] - \frac{K\omega_m}{R_a}$$

Under steady state conditions

$$E = K\omega$$

$$= V - IRa$$

$$T = KIa$$

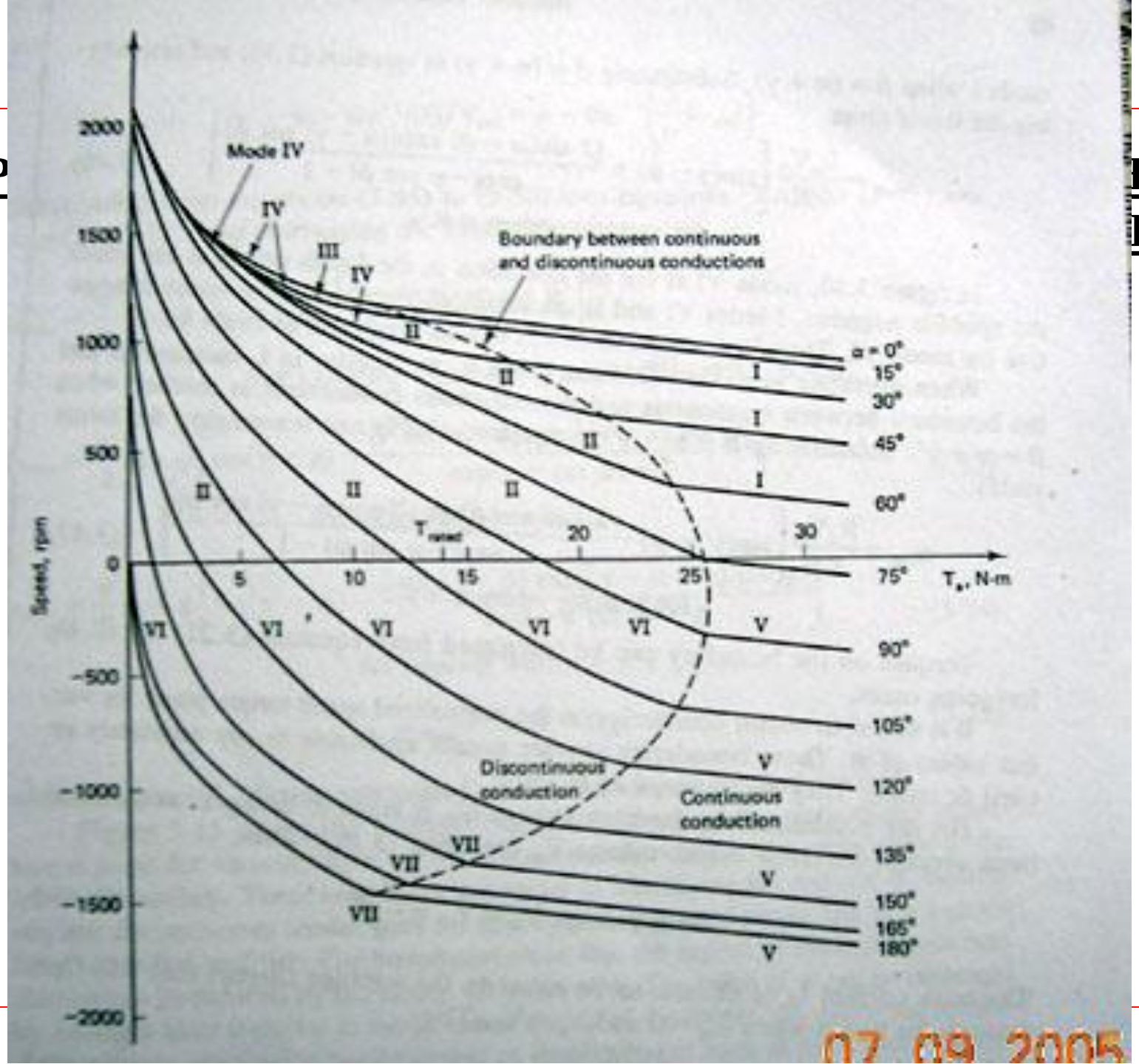
$$\omega = \frac{V}{K} - \frac{T}{K^2} Ra$$

$$\omega = \frac{2V_m}{\pi K} \cos\alpha - \frac{T}{K^2} Ra$$

(Refer example 3.1,3.2 …Power semiconductor controlled drives-G.K. Dubey)

SP **HASE**

E

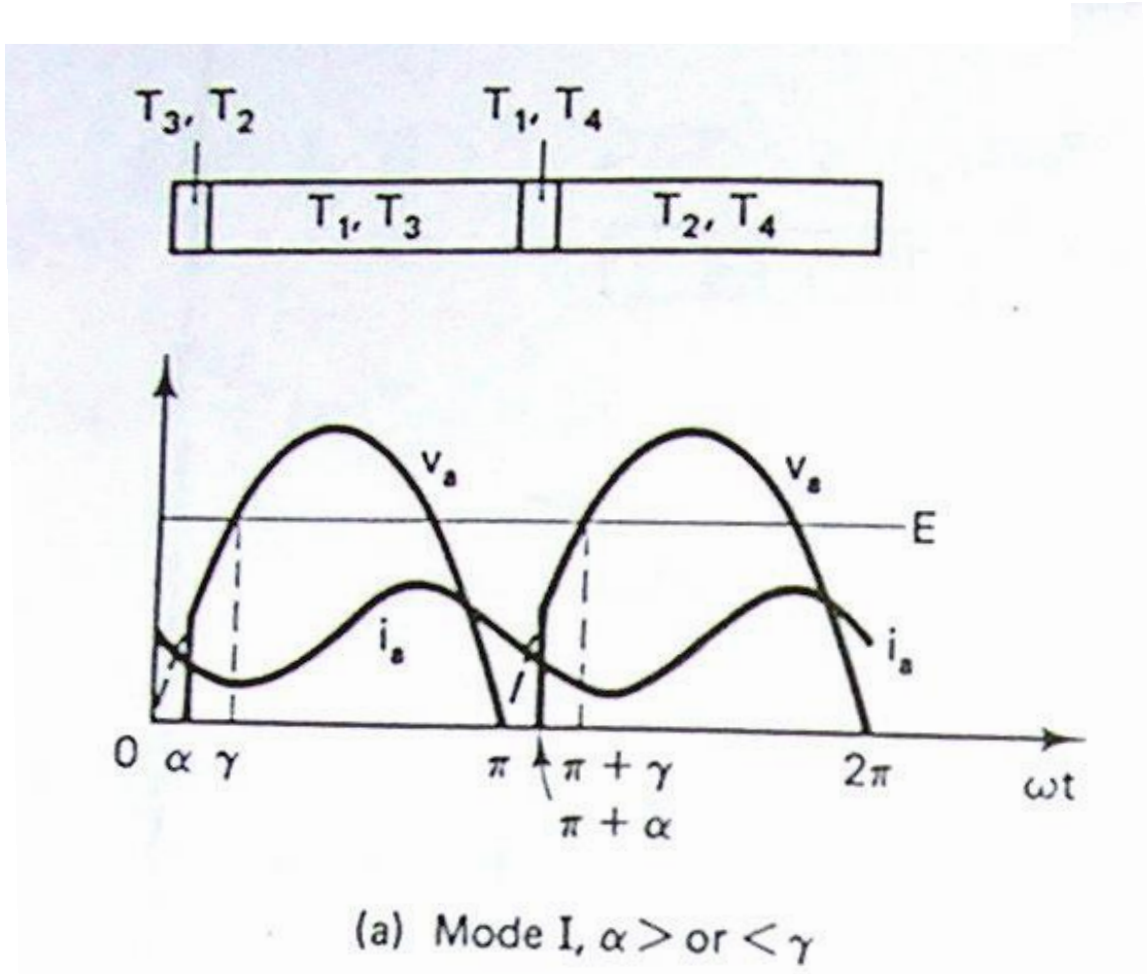


07.09.2005

Rectifier with controlled Flywheeling

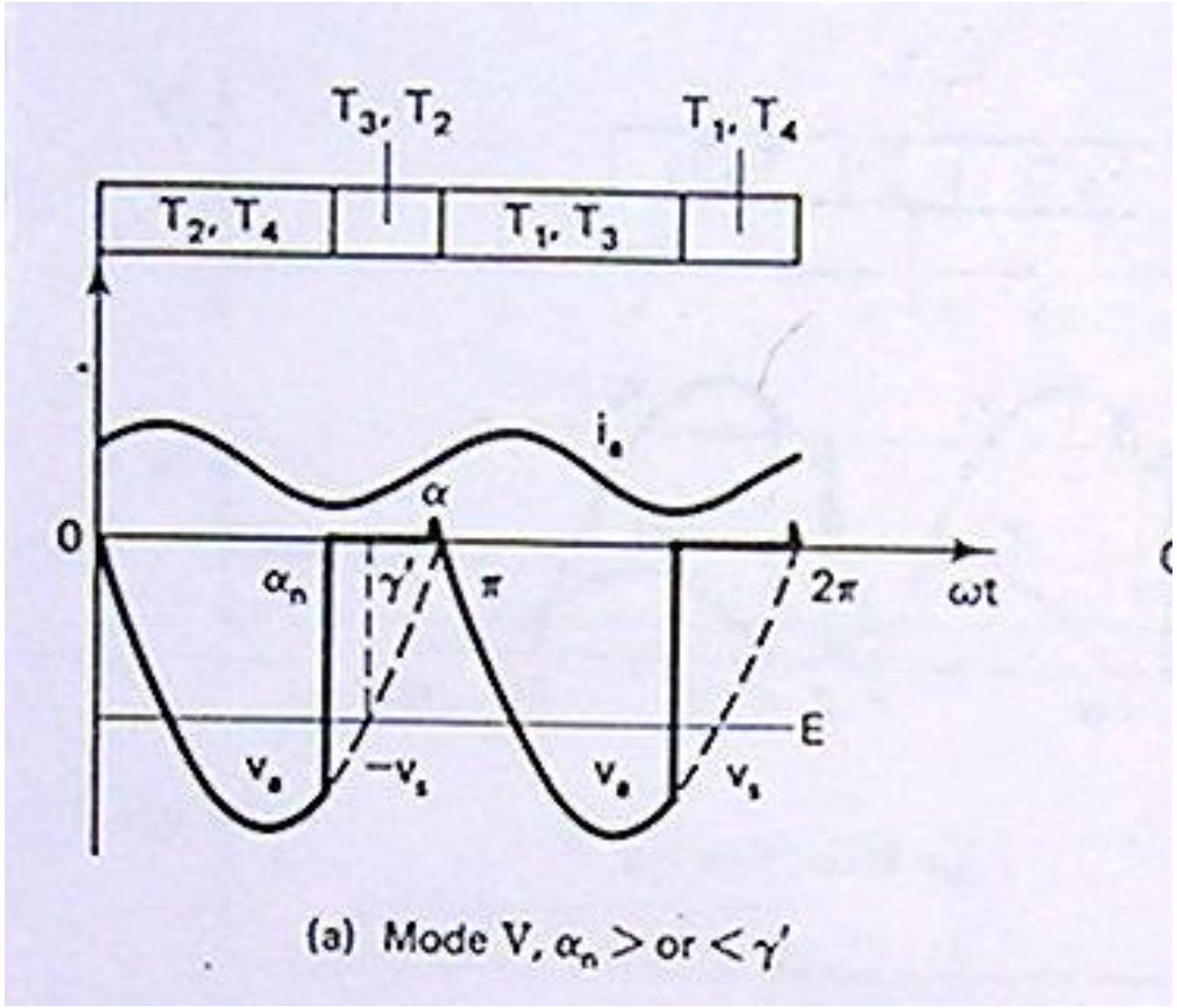
- The output wave form of a single phase controlled rectifier contains positive and negative excursions.
- Current is always positive, hence the +ve voltage mean energy flow from source to load. and vice versa.
- The back and forth power flow indicate reactive energy .(and is due to the negative excursion.)
- It will reduce the power factor.

- During regenerative mode the +ve voltage mean energy flow from source to load. and vice versa.
- In this case the reactive power is due to the +ve excursion.
- In controlled Flywheeling, the -ve excursion during rectification and +ve excursion during inversion are eliminated.
- It is achieved by diverting current to a freewheeling paths formed by series combination of T1,T4 and T2,T3.
- This gives rise to considerable improvements in power factor



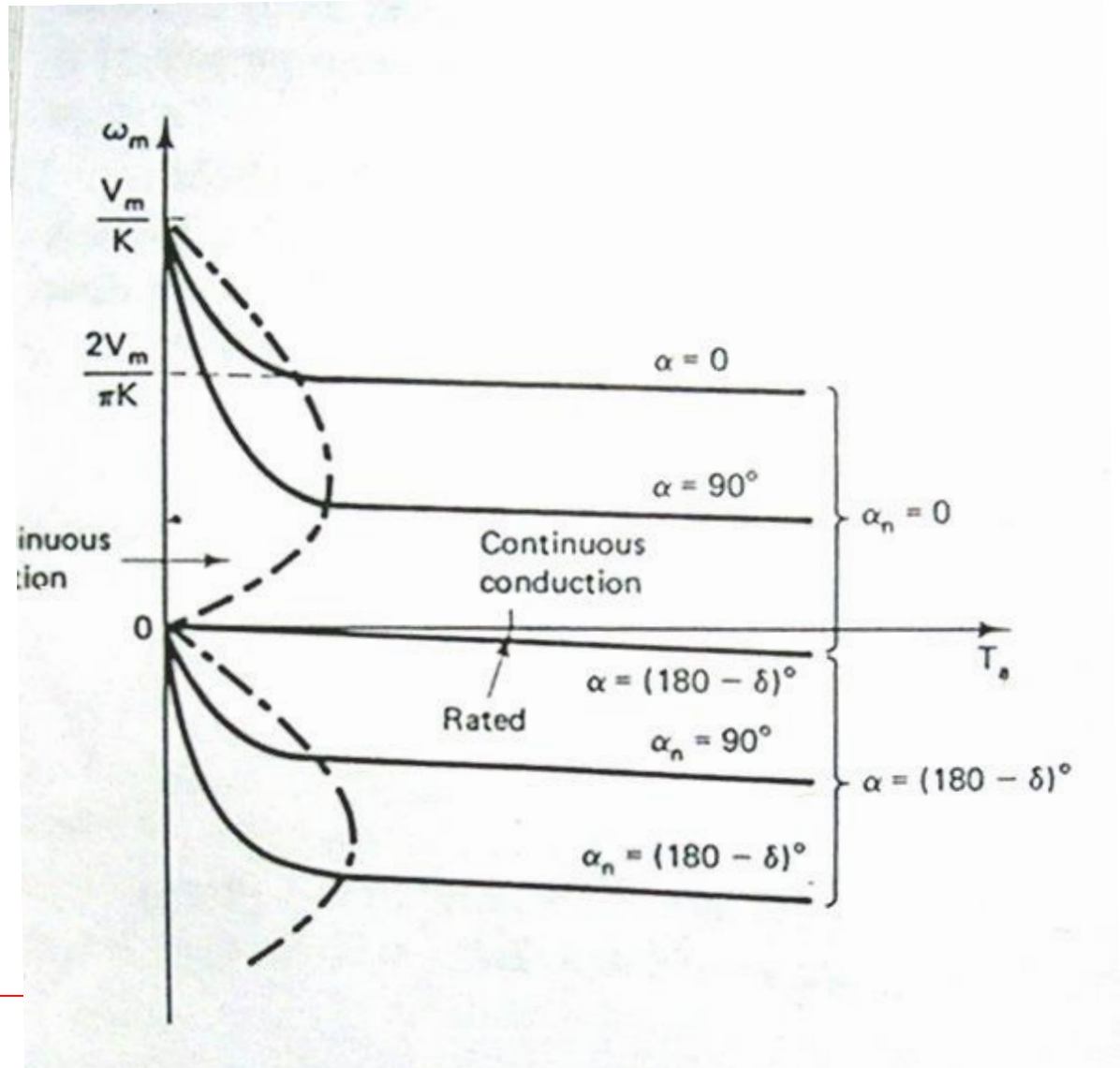
(a) Mode I, $\alpha >$ or $< \gamma$

1Phase rectifier with controlled flywheeling (Motoring)

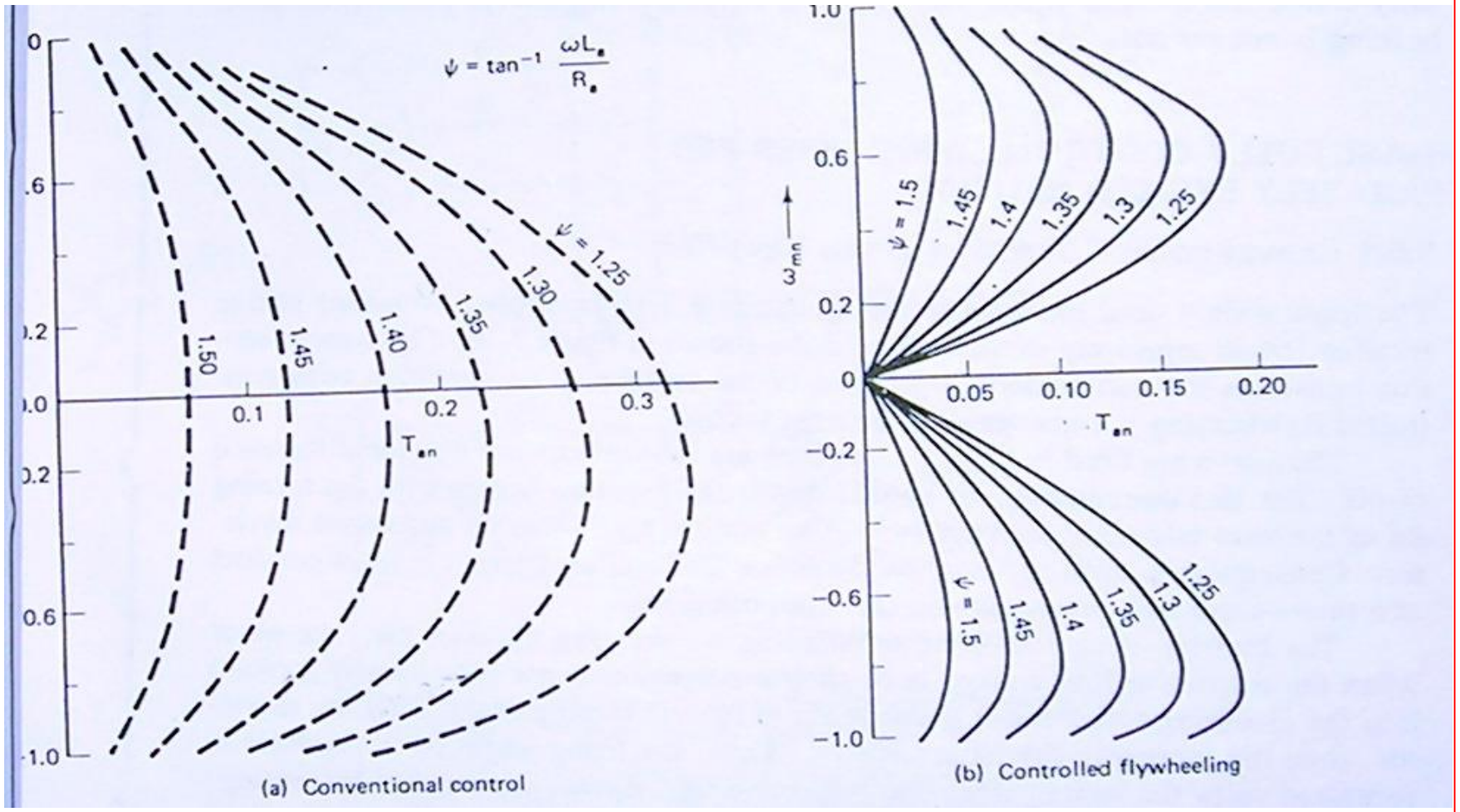


1Phase rectifier with controlled flywheeling(Regeneration)

Speed -Torque characteristic of 1 phase rectifier with controlled flywheeling



Comparison of boundaries for conventional &controlled Flywheeling



SINGLE PHASE DUAL CONVERTER DC DRIVES

SINGLE-PHASE DUAL CONVERTER DRIVE

- Fig. shows a dual converter driving a separately excited dc motor

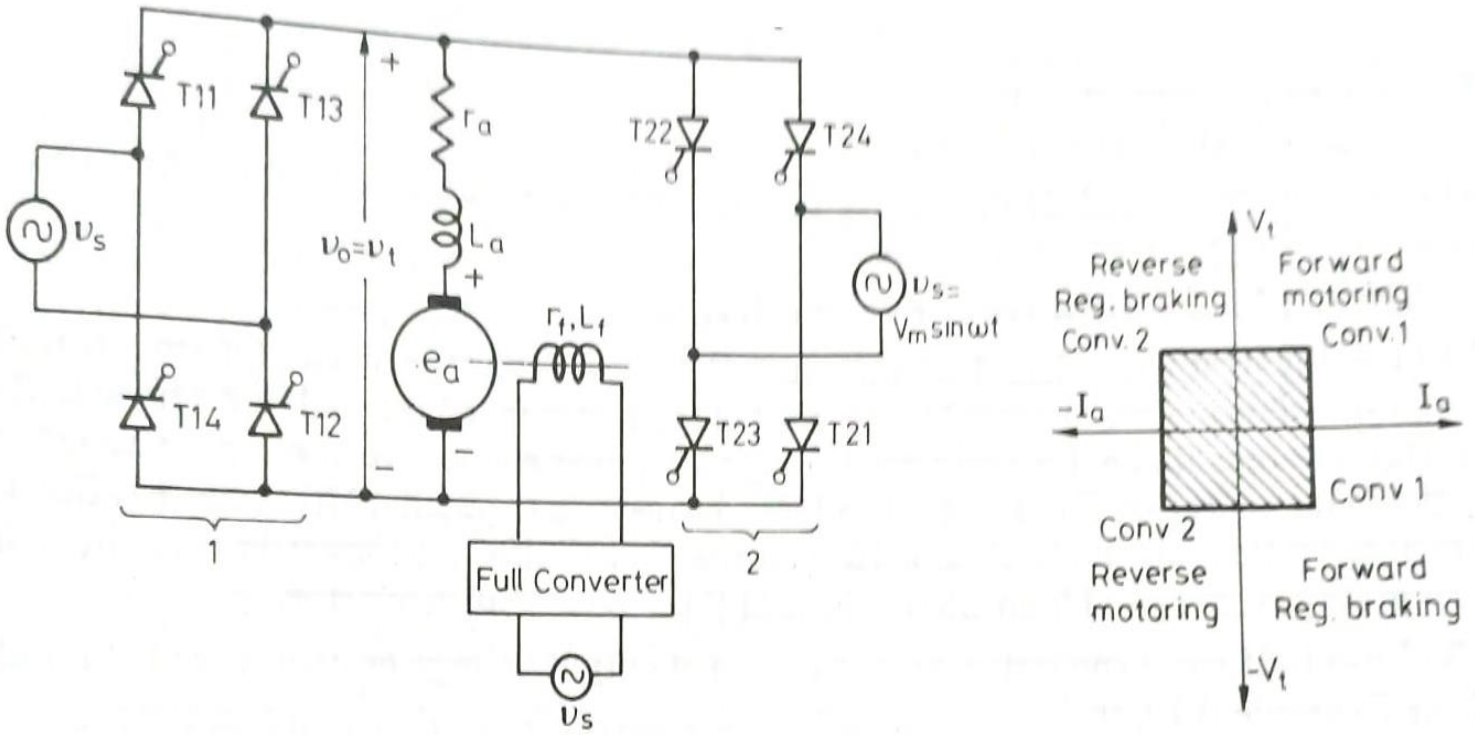


Fig. (a) Single-phase dual converter feeding a separately-excited dc motor (b) four-quadrant diagram.

SINGLE-PHASE DUAL CONVERTER DRIVE

- Dual converter –obtained by connecting 2 full converters in anti-parallel.
- Use is limited to about 15kW drives
- Offers 4-quadrant operation
- For working in first and fourth quadrants, converter 1 is in operation.
- For working in second and third quadrants, converter 2 is energised.
- 4-quadrant operation demands that field winding of the motor is energised from a 1-phase or 3-phase full converter.

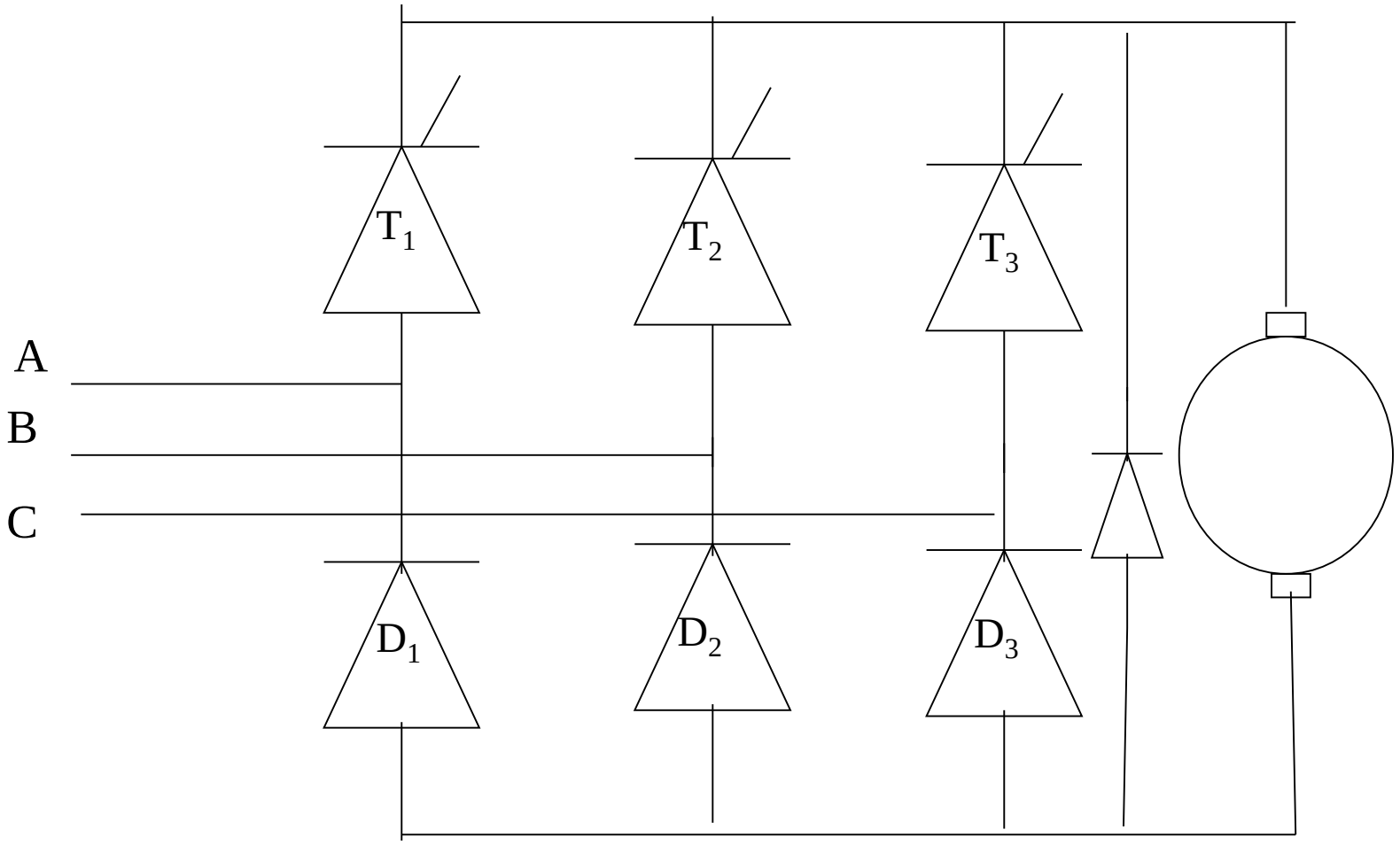
SINGLE-PHASE DUAL CONVERTER DRIVE

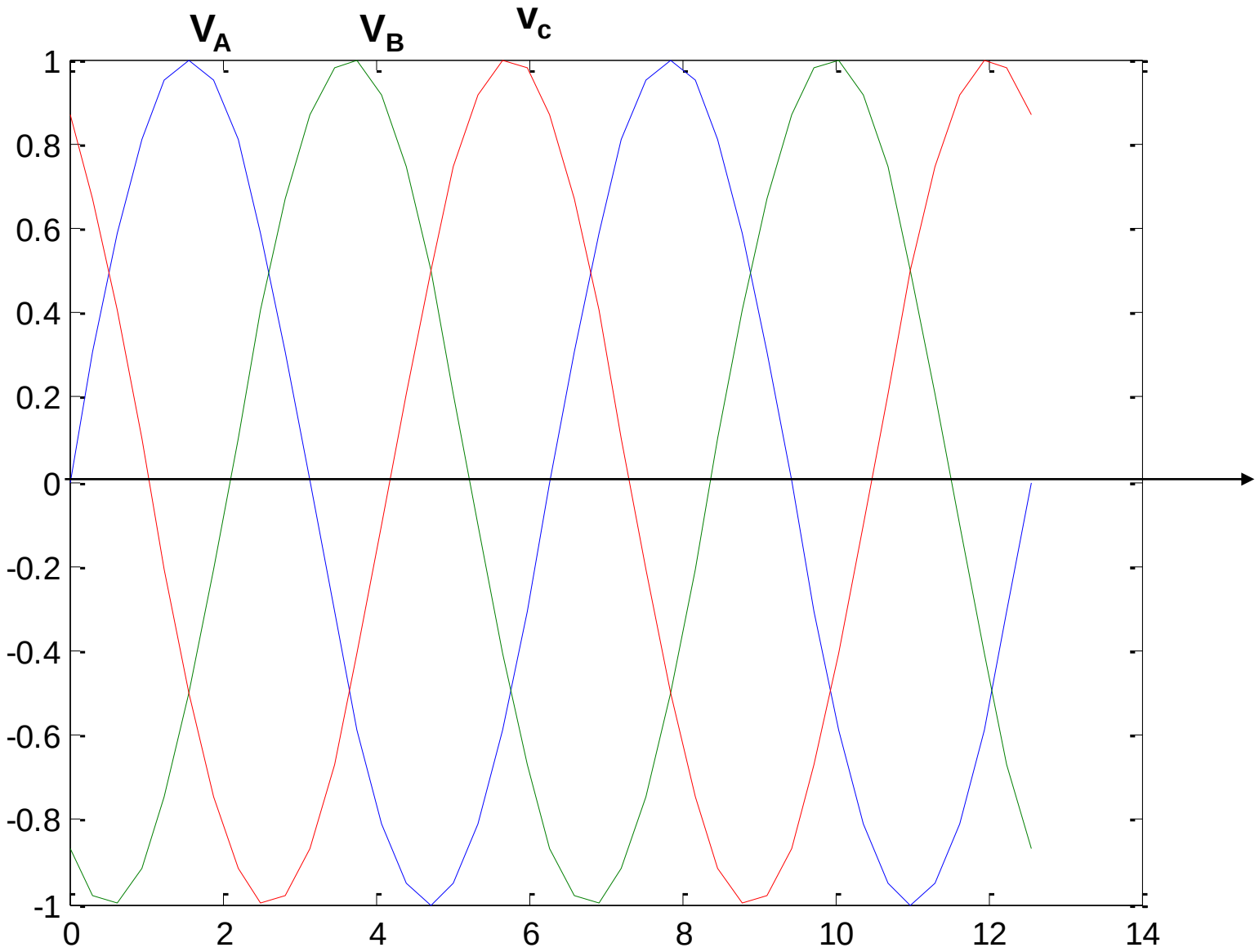
- For converter 1 in operation, $V_t = (2Vm/\pi) \cos \alpha_1$ for $0 \leq \alpha_1 \leq \pi$
- For converter 2 in operation, $V_t = (2Vm/\pi) \cos \alpha_2$ for $0 \leq \alpha_2 \leq \pi$

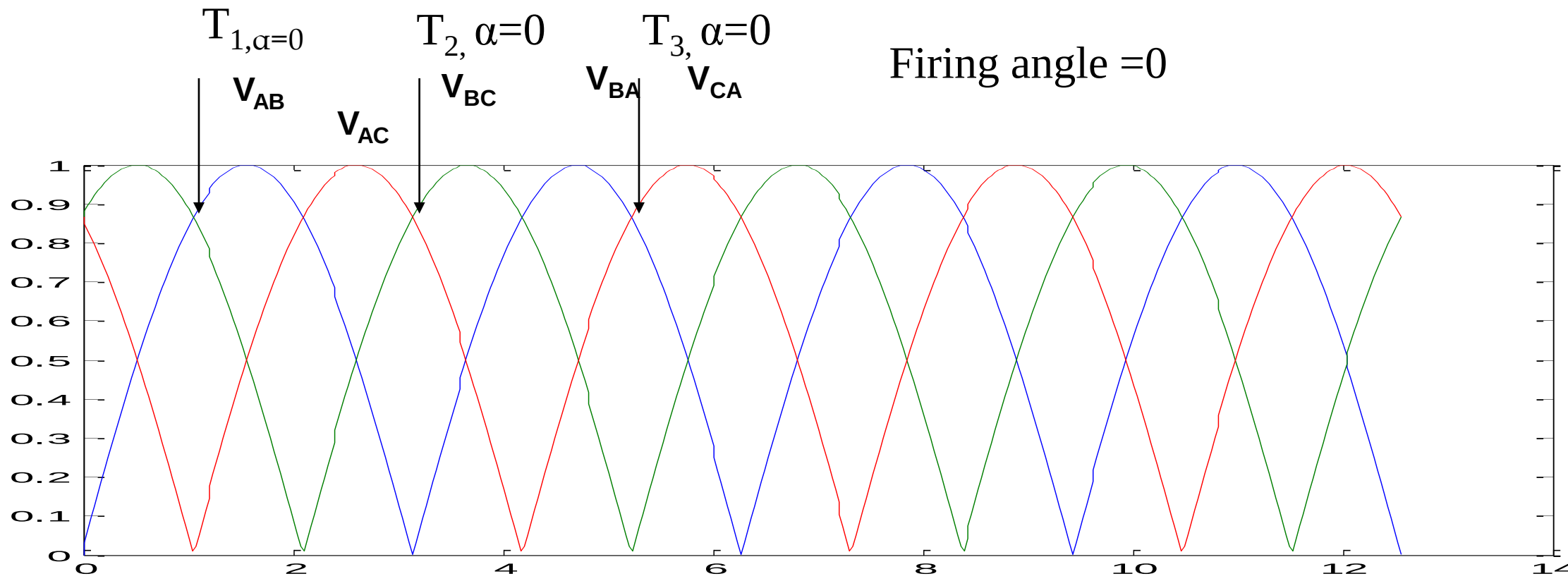
  where $\alpha_1 + \alpha_2 = \pi$
- For the field converter, $V_f = (2Vm/\pi) \cos \alpha_3$ for $0 \leq \alpha_3 \leq \pi$
- Converter 1 with $\alpha_1 < 90^o$ operates the motor in forward motoring mode in quadrant 1
- Converter 1 with $\alpha_1 > 90^o$ and with field excitation reversed operates the motor in forward regenerative braking mode in quadrant 4
- Converter 2 with $\alpha_2 < 90^o$ operates the motor in reverse motoring mode in quadrant 3
- Converter 2 with $\alpha_2 > 90^o$ and with field excitation reversed operates the motor in reverse regenerative braking mode in quadrant 2

Three Phase Half Controlled Rectifiers.

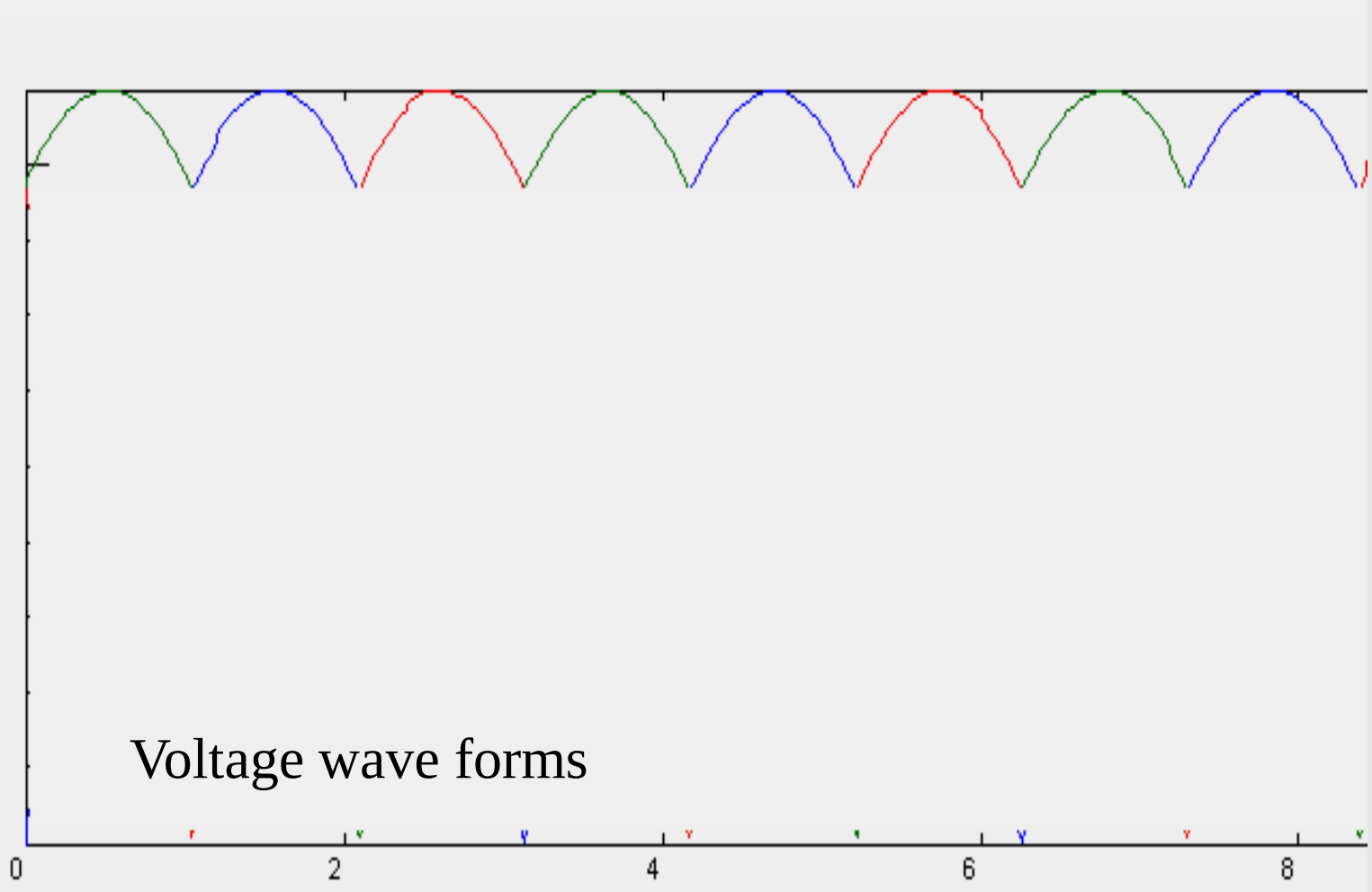
Half Controlled Rectifier -3 thyristors-3Diodes
One Diode is used as Free Wheeling Diode
Action is similar to that of single-phase rectifiers.


A
B
C
T1
T2
T3
D1
D2
D3

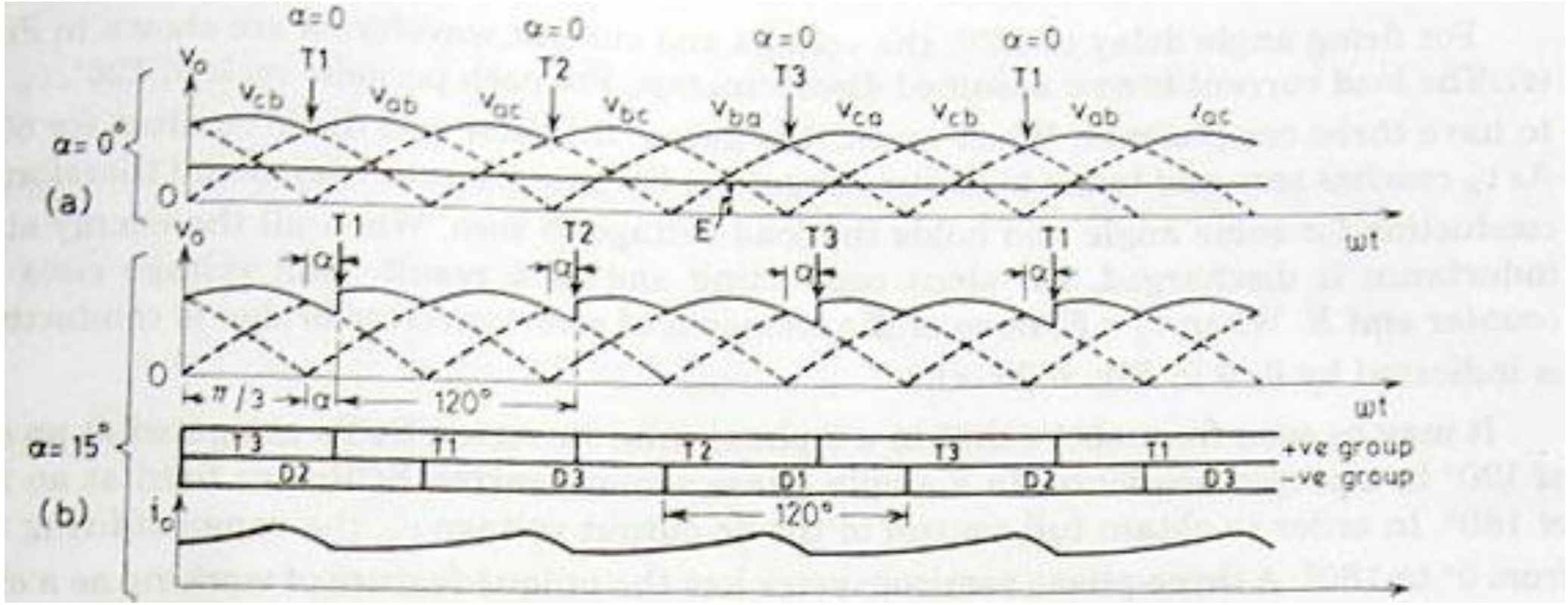

V_A
V_B
V_c
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
2
4
6
8
10
12
14



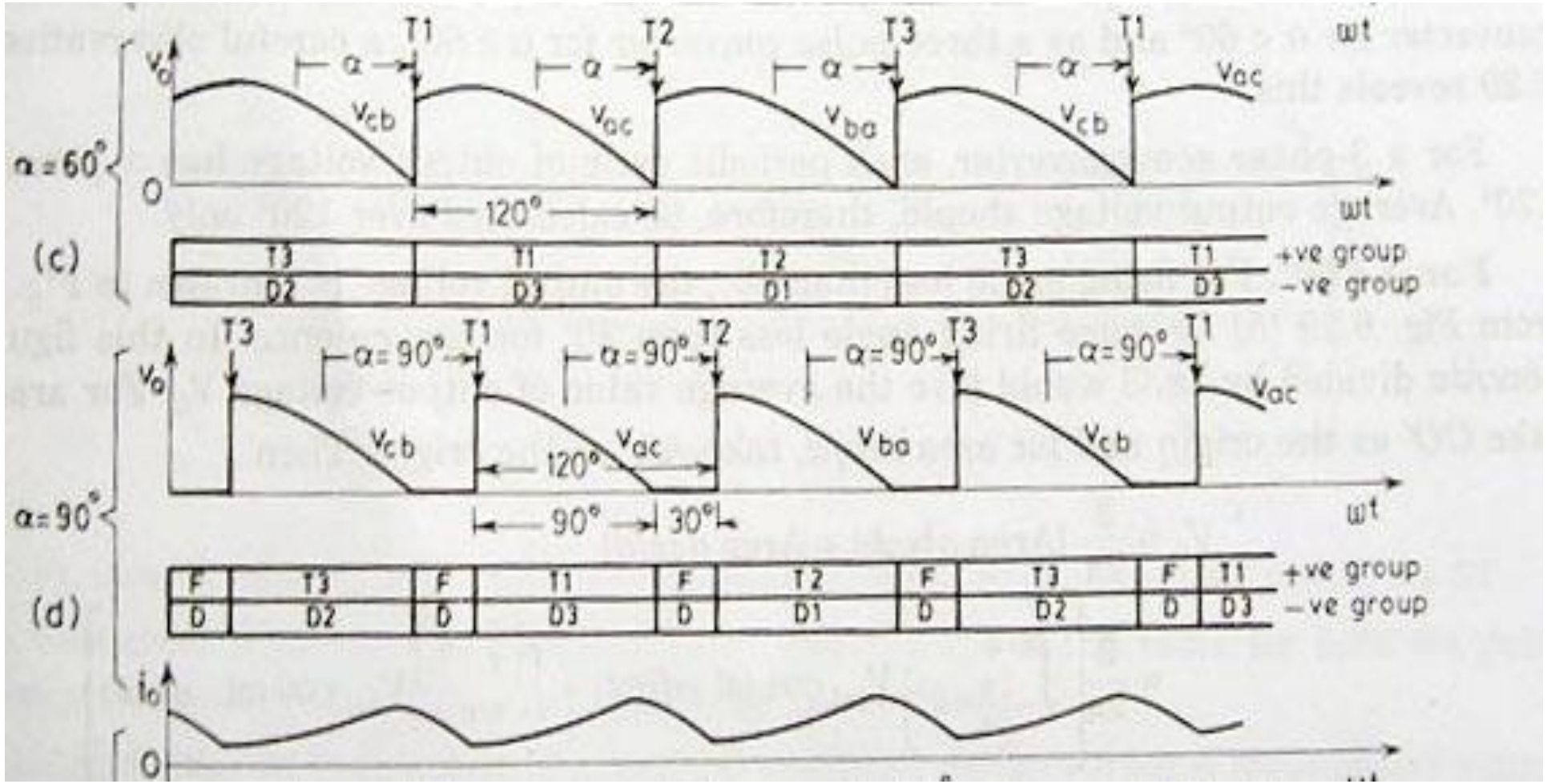
T_3	T_1	T_2	T_3	T_1	
D_2	D_3	D_1	D_2		



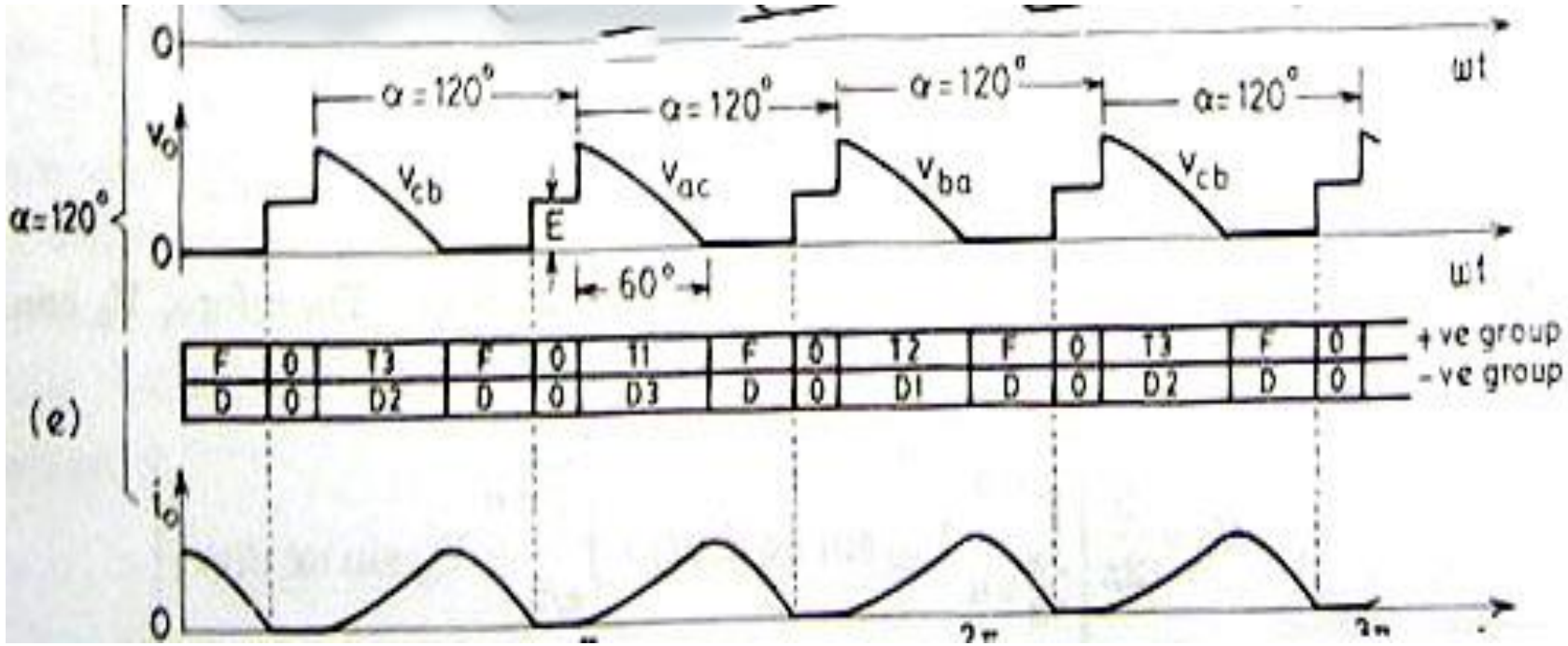
Voltage wave forms



Firing Angle =15^0 Current Continuous



Current Continuous, T_3 becomes off due to natural commutation


α=120°
(e)
0
ωt
α = 120°
α = 120°
α = 120°
α = 120°
v_o
V_{cb}
V_{ac}
V_{ba}
V_{cb}
E
60°
0
ωt
F
D
T3
D2
T1
D3
T2
D1
T3
D2
+ ve group
- ve group
i_o
0
2π
3π

Average Voltage

For $\alpha<60^0$

$$V_0 = \frac{1}{2\pi/3}\left[\int_{\frac{\pi}{3}+\alpha}^{2\pi/3} V_{ml}\sin\varpi t d(\varpi t) + \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}+\alpha} V_{ml}\sin\varpi t d(\varpi t)\right]$$

$$= \frac{3V_{ml}}{2\pi}[1+\cos\alpha]$$

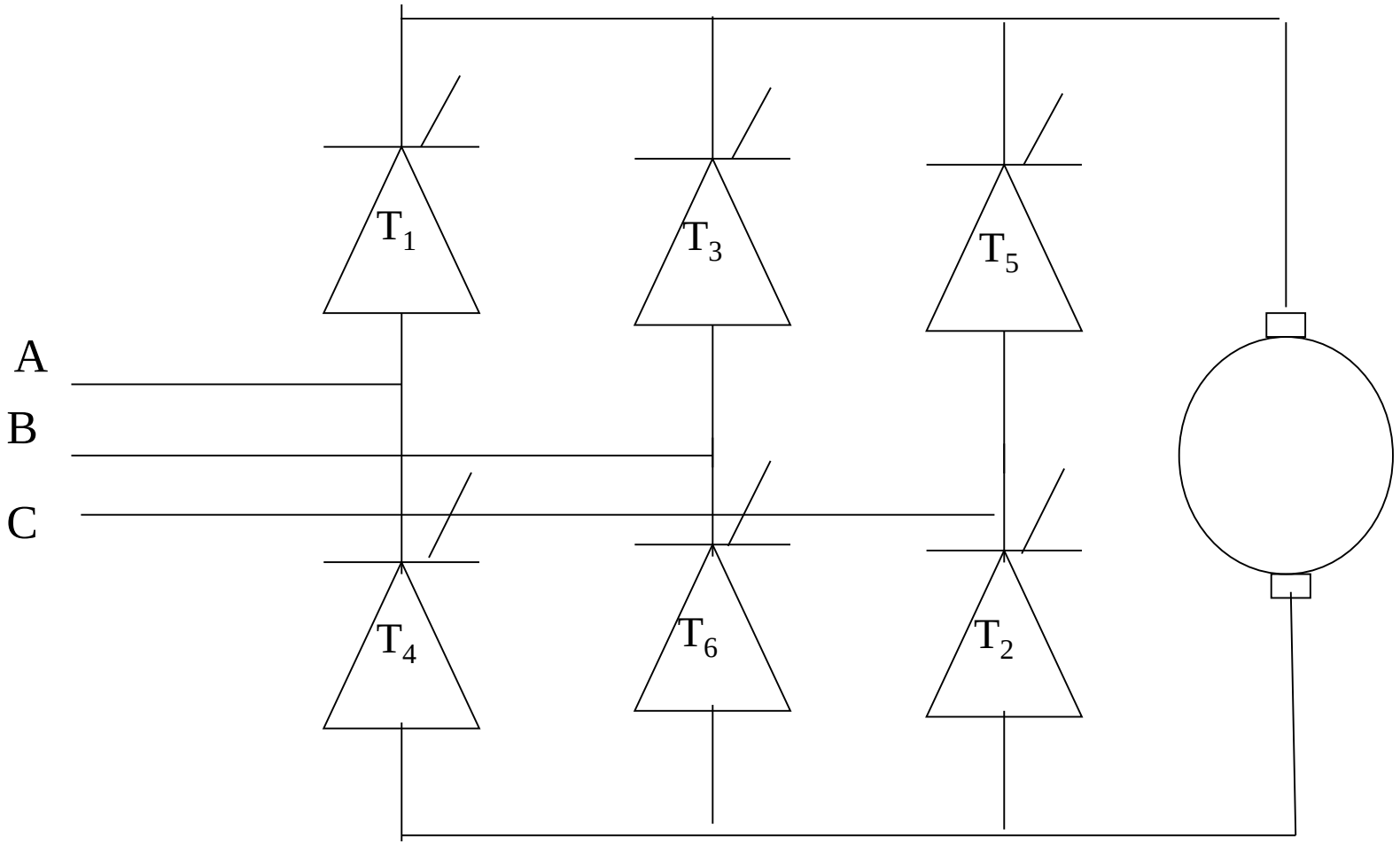
For $\alpha >= 60^0$

$$V_0 = \frac{3}{2\pi}\int_{\alpha}^{\pi} V_{ml} Sin\varpi t d(\varpi t)$$

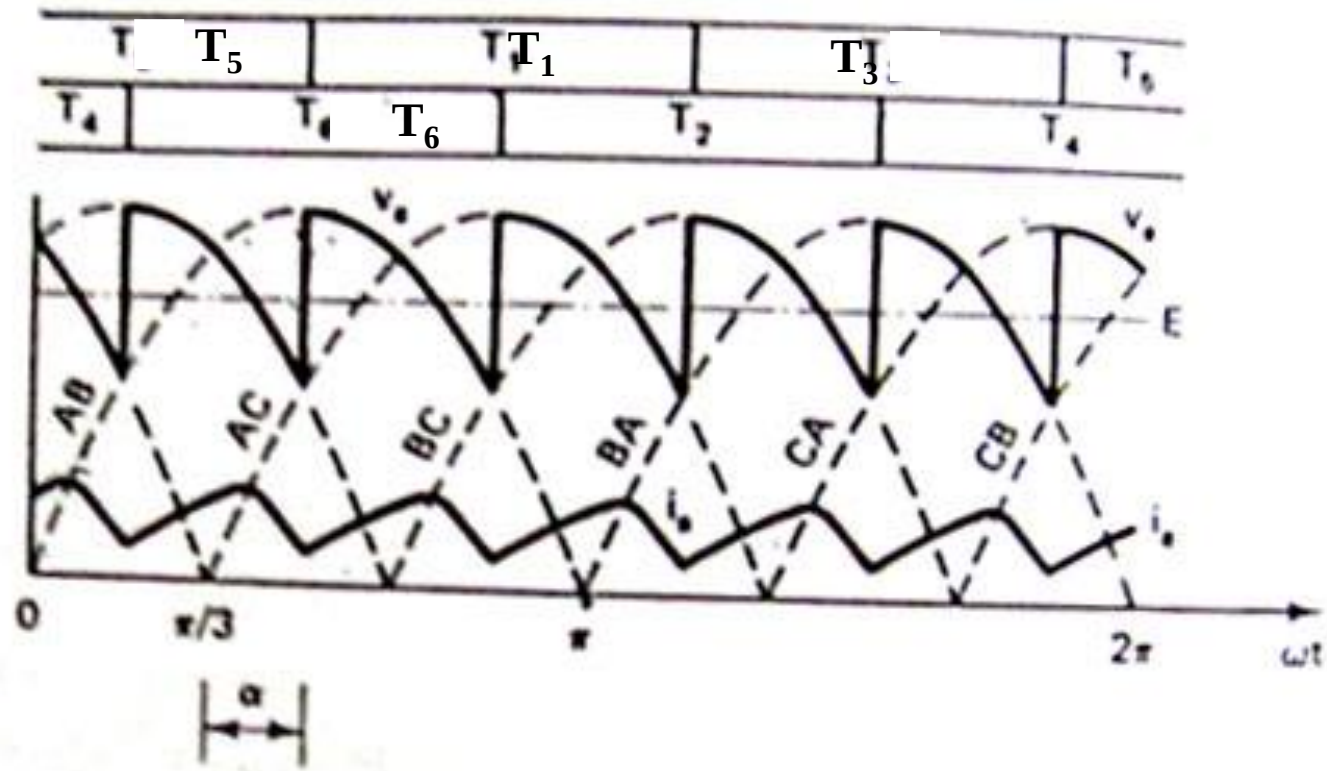
$$= \frac{3}{2\pi}[1+\cos\alpha]$$

Hence the average voltage is the same for both the cases

Three Phase Full Converter



(a) Mode I



Mode -1

$$V_0 = \frac{1}{\pi/3}\left[\int_{\frac{\pi}{3}+\alpha}^{\frac{2\pi}{3}+\alpha} V_{ml}\sin\varpi t\, d(\varpi t)\right]$$

$$= \frac{3V_{ml}}{\pi}\cos\alpha$$

$$E = K\omega$$

$$= V - IRa$$

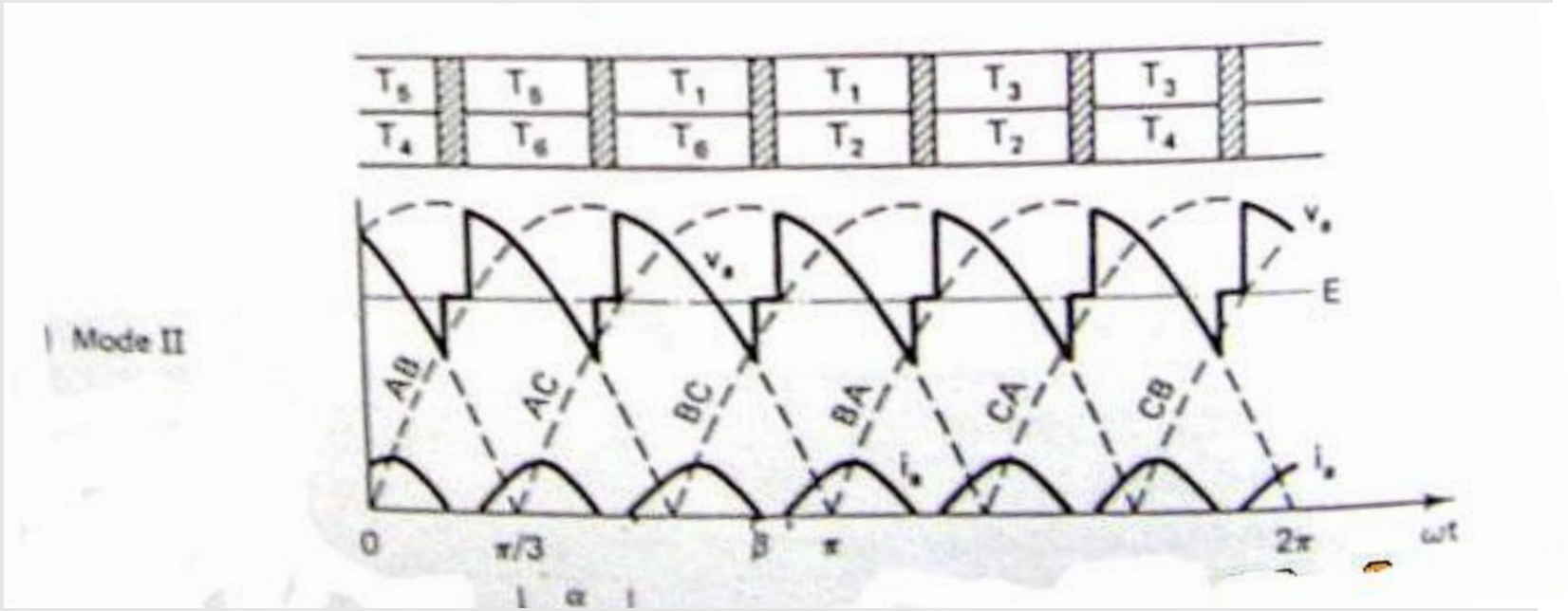
$$T = KIa$$

$$\omega = \frac{V}{K} - \frac{T}{K^2} Ra$$

$$\omega = \frac{3V_{ml}}{\pi K} \cos\alpha - \frac{T}{K^2} Ra$$

- In mode 1thyristors conduct for120^0
- Thyristors are triggered in the order of their no.
- T1,T2,T3 etc are triggered with a phase difference of $60^{0.}$
- The transfer of current from an outgoing Thyristor to an in coming takes place when the respect line voltage forward biases incoming one and reverse biases the other.

MODE-II


Mode II
T5
T6
T1
T1
T3
T3
T4
T6
T6
T2
T2
T4
E
AB
AC
BC
BA
CA
CB
0
π/3
β
π
2π
ωt
α

In mode -II

$$V_a = L_a \frac{di}{dt} + R_a i_a + K\omega_m = V_m \sin\omega t, \ldots\ldots when\ \alpha + \frac{\pi}{3} \le \omega t \le \beta$$

$$V_a = E \ldots\ldots when \beta \le \omega t \le \alpha + \frac{2\pi}{3}$$

MODE-II

Current through drive consists of

- component due to ac source
- component due to back emf
- transient component

$$i_a(\omega t) = \frac{V_m}{Z}\sin(\omega t - \psi) - \frac{K\omega_m}{R_a} + k_1 \exp(\frac{-t}{\tau_a})$$

$$Z = [R_a{}^2 + (\omega La)^2]^{1/2}$$

$$\tau_a = \frac{L_a}{R_a}$$

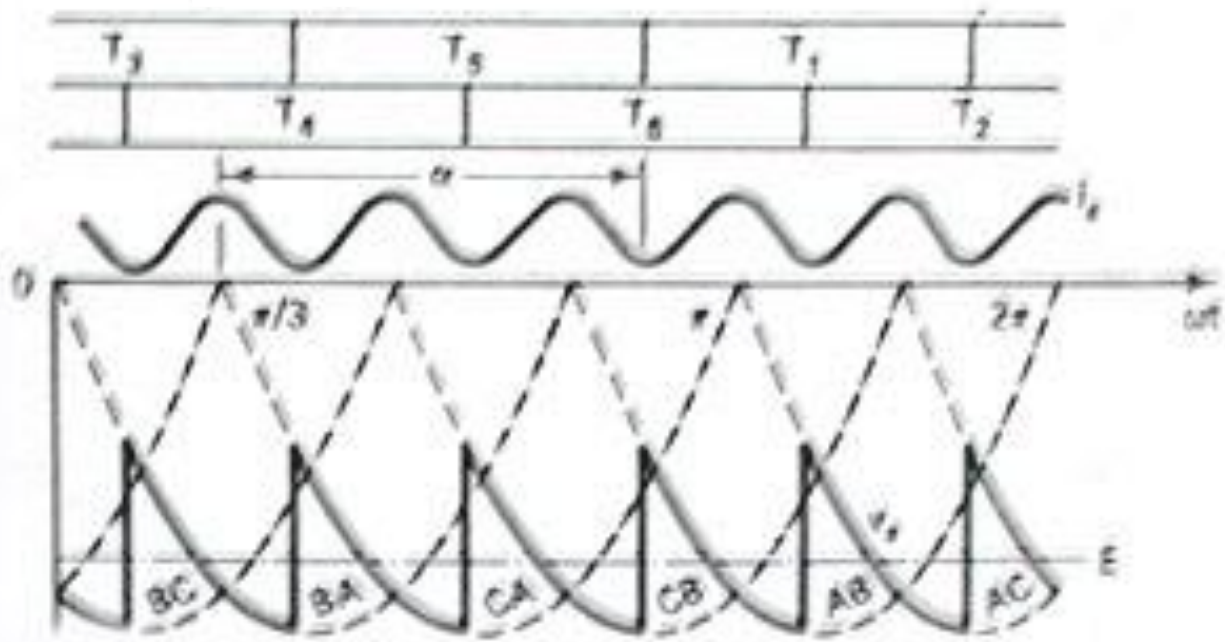
$$\psi = \tan^{-1}\frac{\omega L_a}{R_a}$$

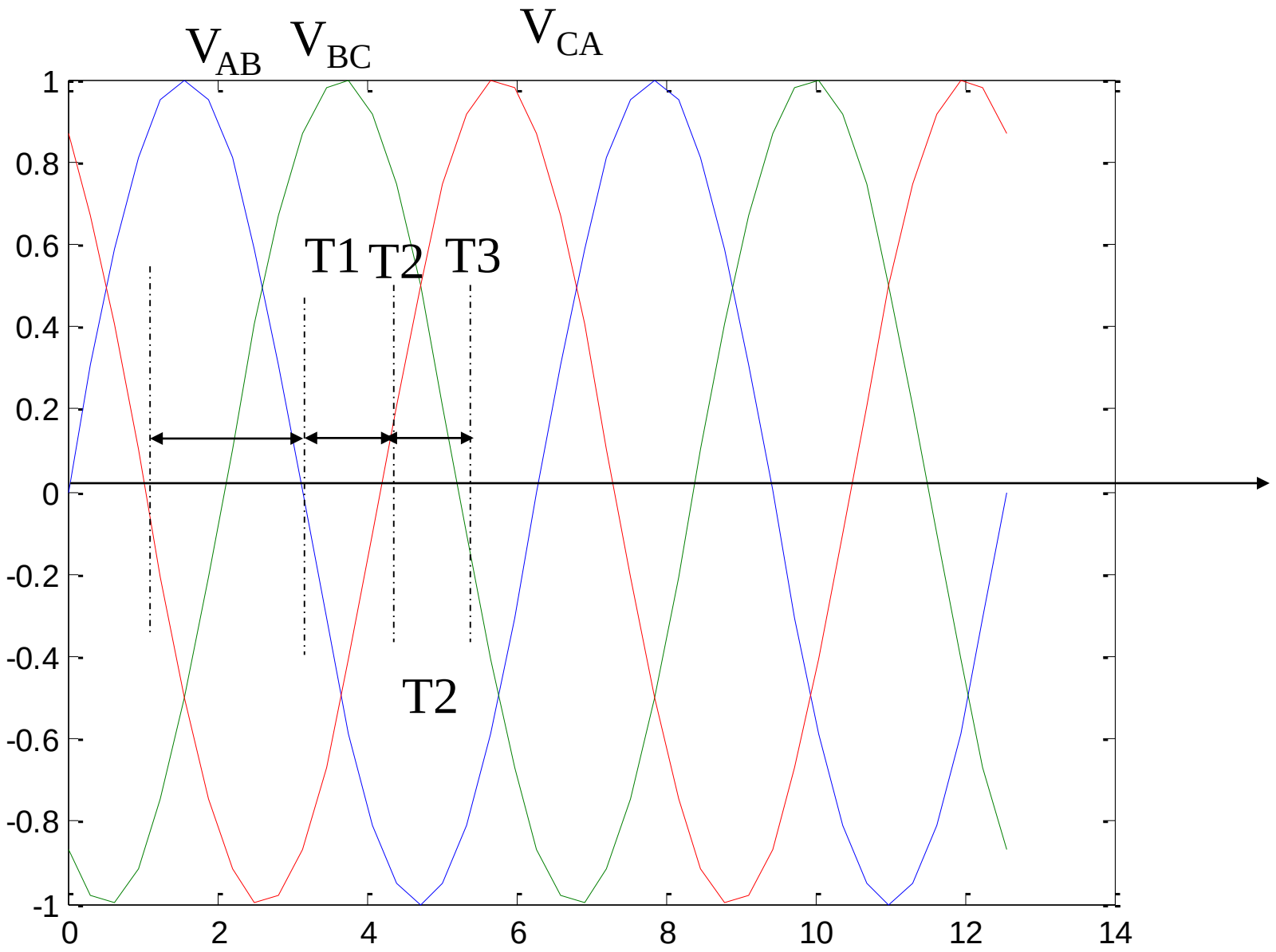
$$i_a(\omega t) = \frac{Vm}{Z}\left[\sin(\omega t - \psi) - \sin(\alpha + \frac{\pi}{3} - \psi)\exp\{(\alpha + \frac{\pi}{3} - \omega t)\cot\psi\}\right]$$

$$-\frac{K\omega_m}{R_a}\left[1 - \exp\{(\alpha + \frac{\pi}{3} - \omega t)\cot\psi\}\right]$$

Put wt=Bhence we can find the point of discontinuity.

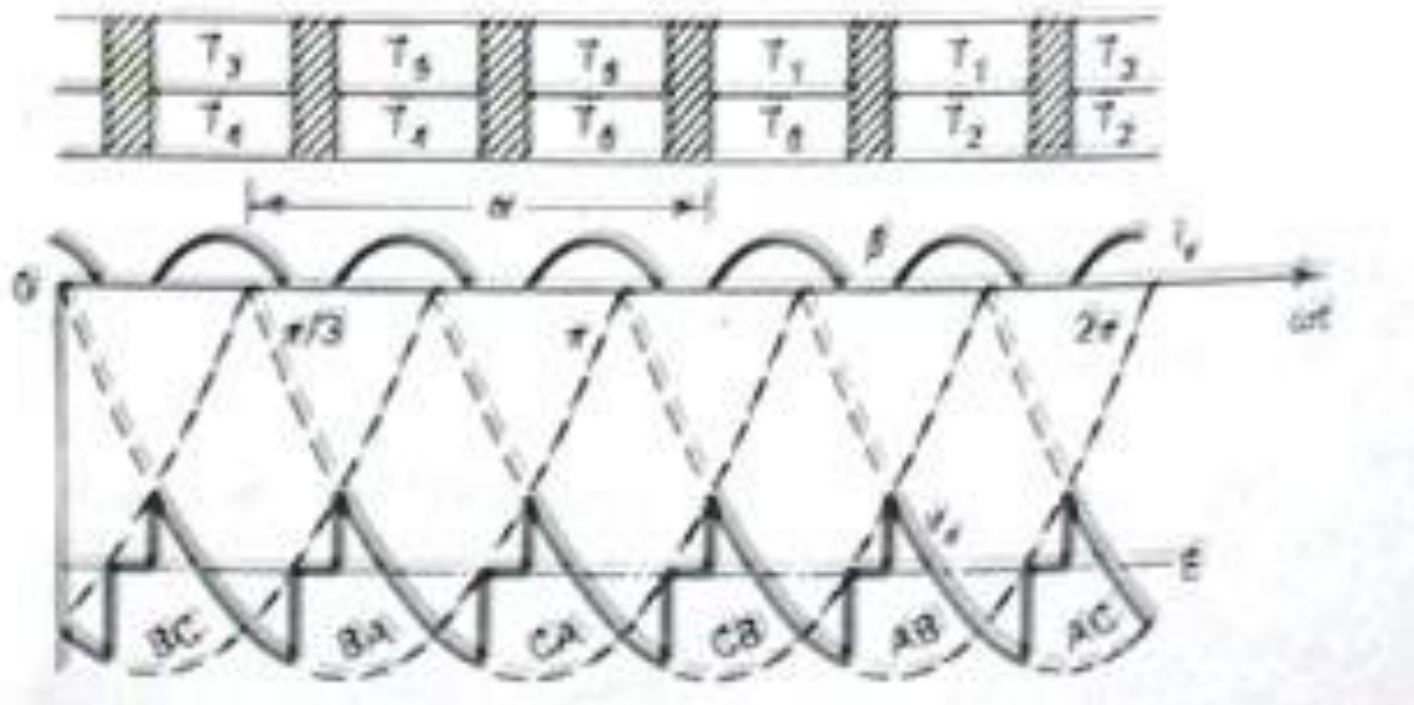
(c) Mode III



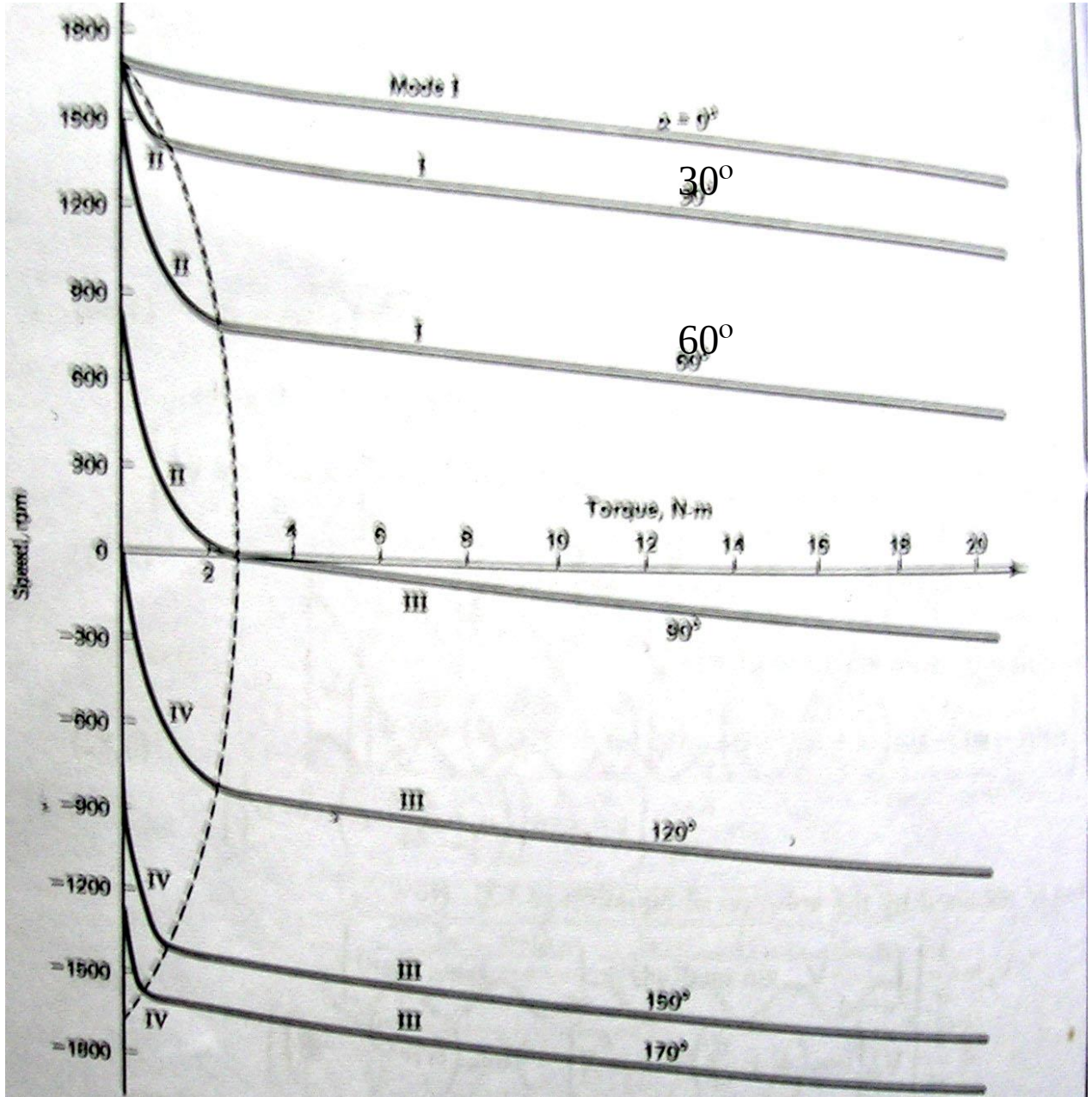

V_{AB}
V_{BC}
V_{CA}
T1
T2
T3
T2
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
2
4
6
8
10
12
14

T1
T2
T3
0
2
4
6
8
10
12
14
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

(d) Mode IV



SPEED TORQUE CURVE OF A THREE PHASE FULLY CONTROLLED RECTIFIER DRIVE

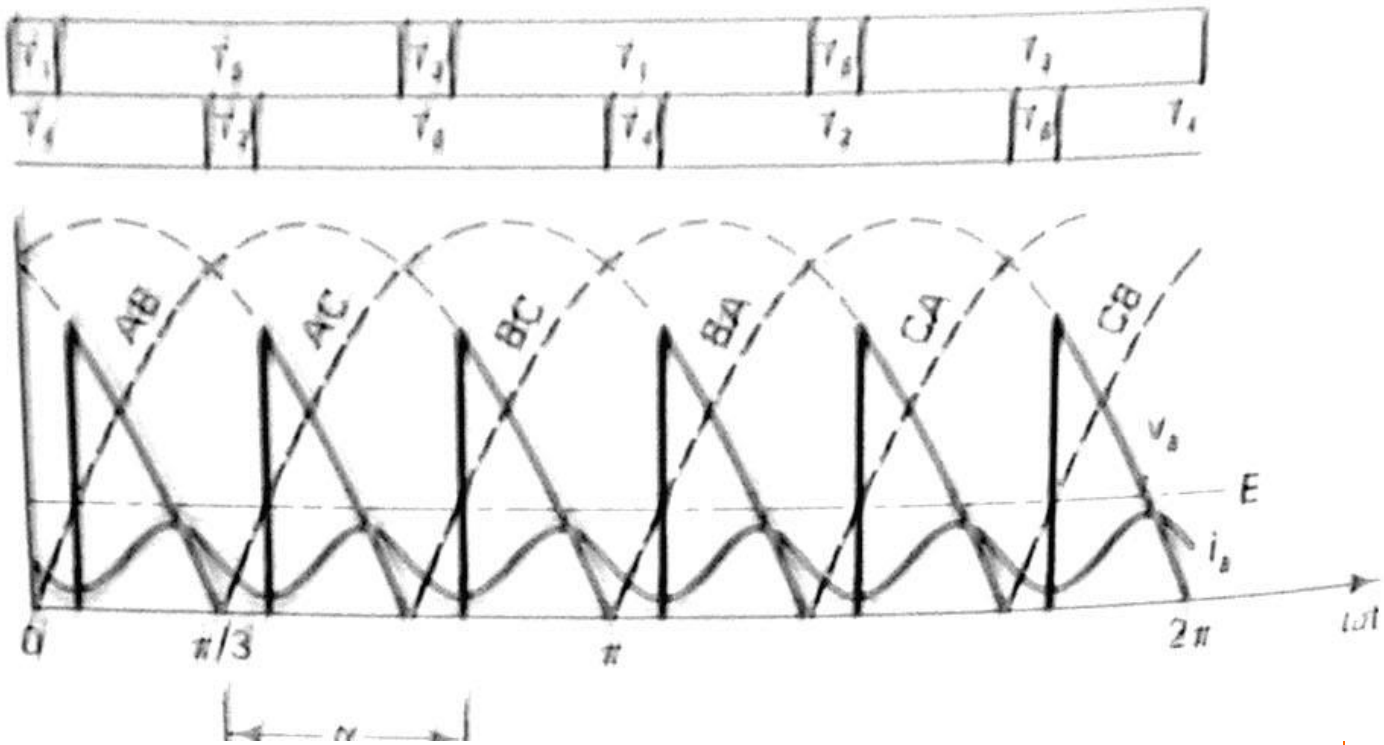


Controlled flywheeling –Motoring and Regeneration

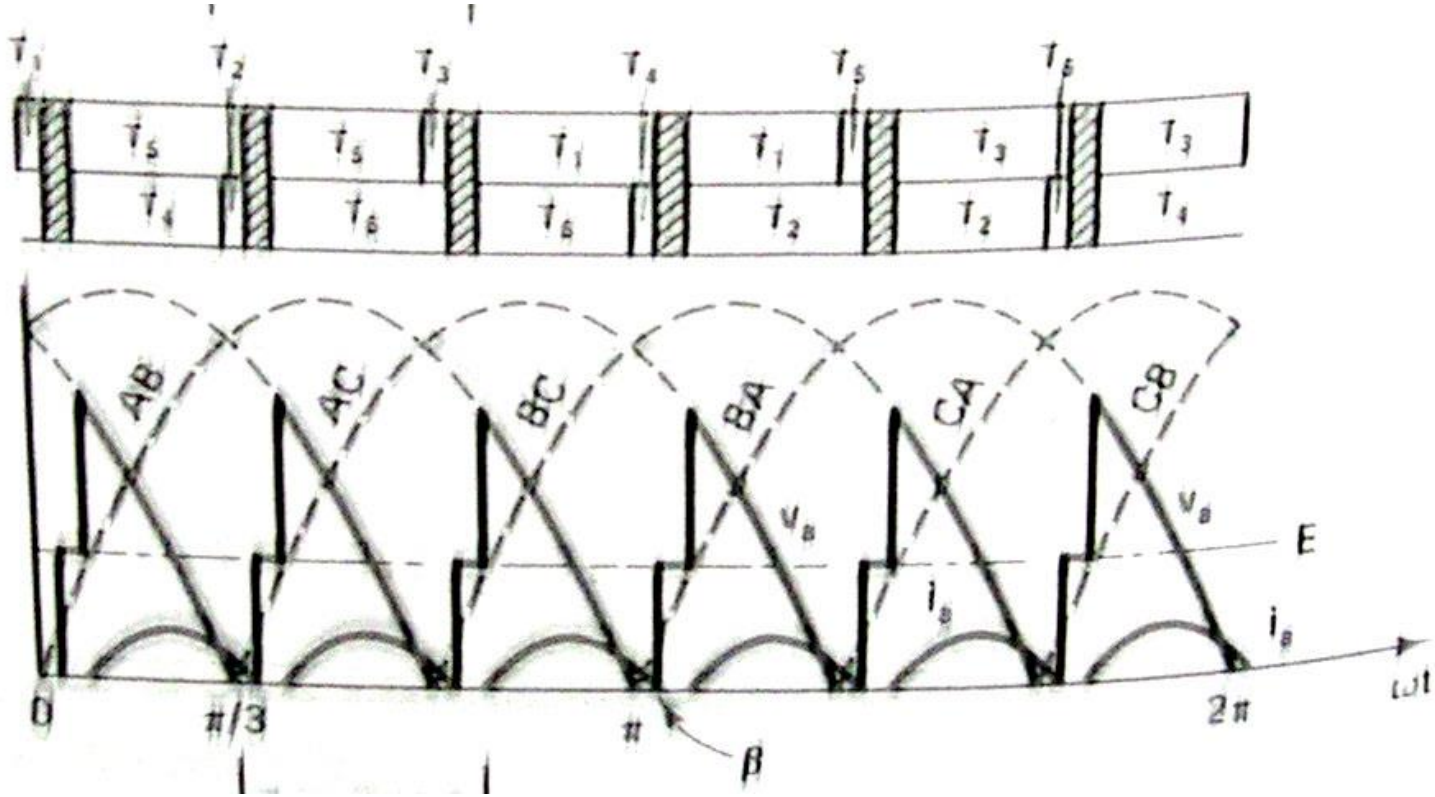
- The output wave form of a single/three phase controlled rectifier contains positive and negative excursions.
- Current is always positive, hence the +ve voltage mean energy flow from source to load. and vice versa.
- The back and forth power flow indicate reactive energy .(and is due to the negative excursion.)
- It will reduce the power factor.

- During regenerative mode the +ve voltage mean energy flow from source to load. and vice versa.
- In this case the reactive power is due to the +ve excursion.
- In controlled Flywheeling, the -ve excursion during rectification and +ve excursion during inversion are eliminated.
- It is achieved by diverting current to a freewheeling paths formed by series combination of thyristors.
- This gives rise to considerable improvements in power factor

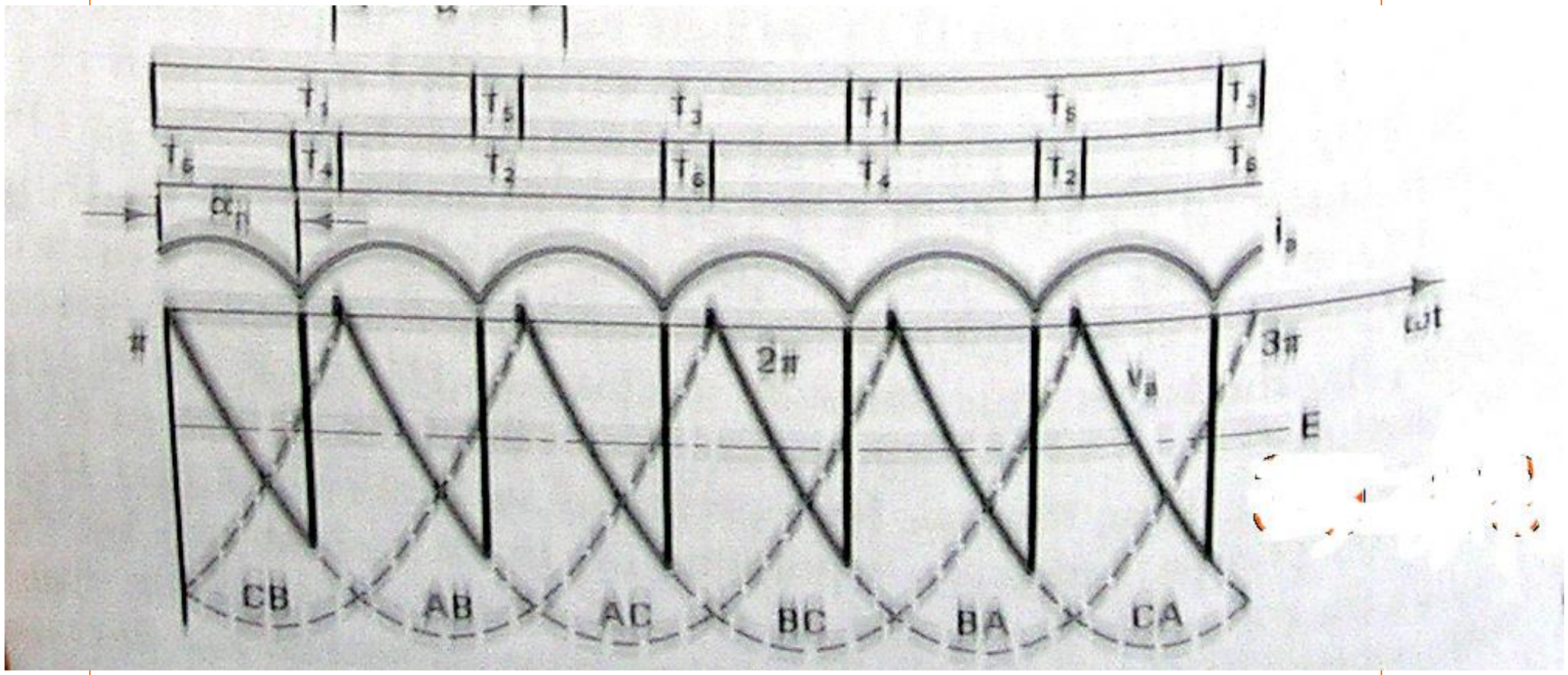
(a) Mode V



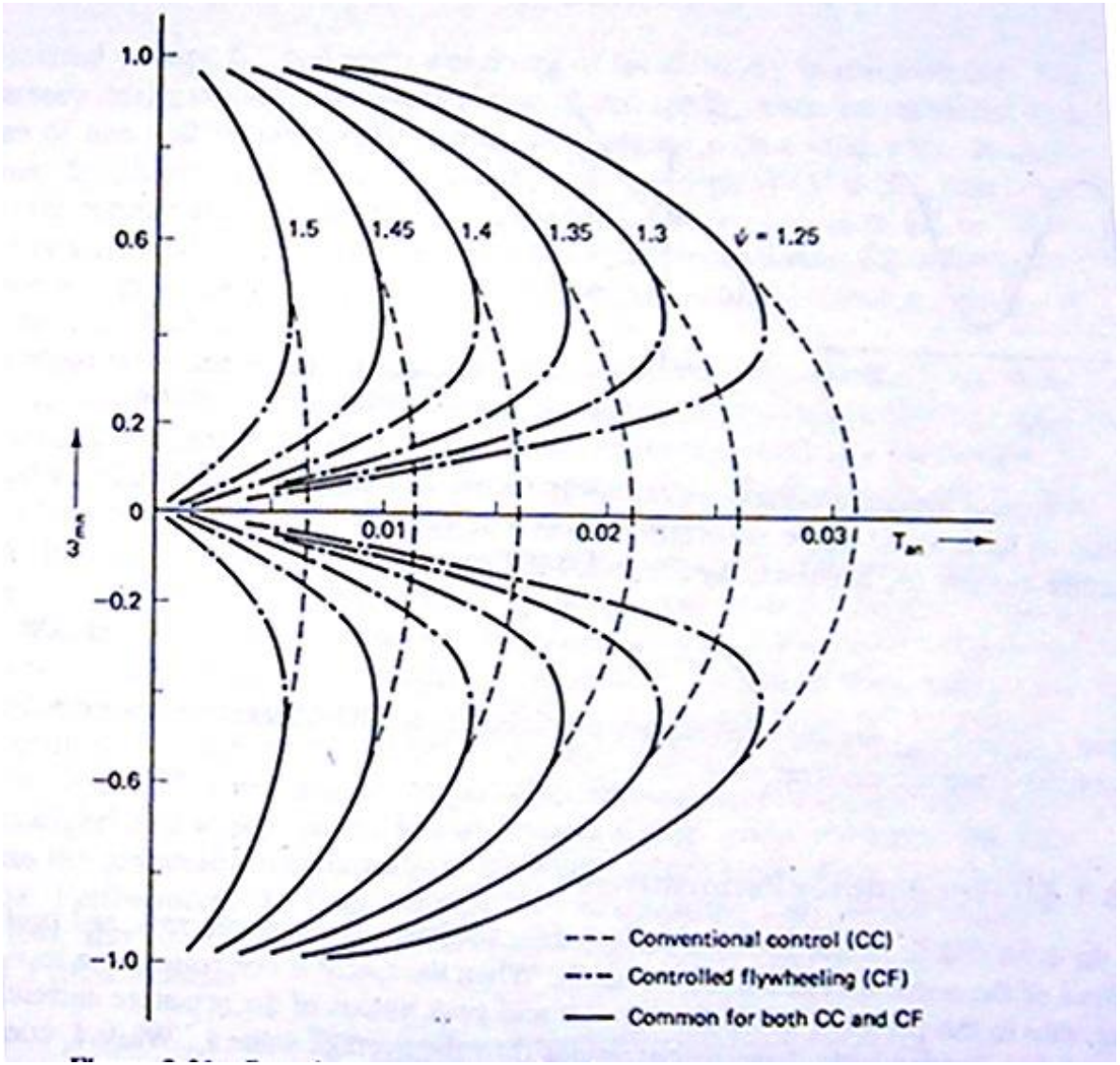
(b) Mode VI


T1
T2
T3
T4
T5
T6
T5
T4
T5
T6
T1
T6
T1
T2
T3
T2
T3
T4
AB
AC
BC
BA
CA
CB
vB
vB
E
iB
iB
0
π/3
π
β
2π
ωt

MODE-VII


T1
T5
T3
T1
T5
T3
T6
T4
T2
T6
T4
T2
T6
α_{rr}
i_a
π
2π
3π
ωt
V_a
E
CB
AB
AC
BC
BA
CA

BOUNDERY CURVES-CONTINUOUS,DISCONTINUOUS MODE WITH FLYWHEELING(3PHASE RECTIFIER)



CHOPPER FED DC DRIVES

Choppers are basically DC to DC converters.
Chopper can be built from any devices such as thyristors, MOSFET, GTO, Power transistor etc..

Why Chopper circuits?

High efficiency.
Flexibility in control.
Light weight.
Small size.
Quick response.
Regeneration down to very low speed.

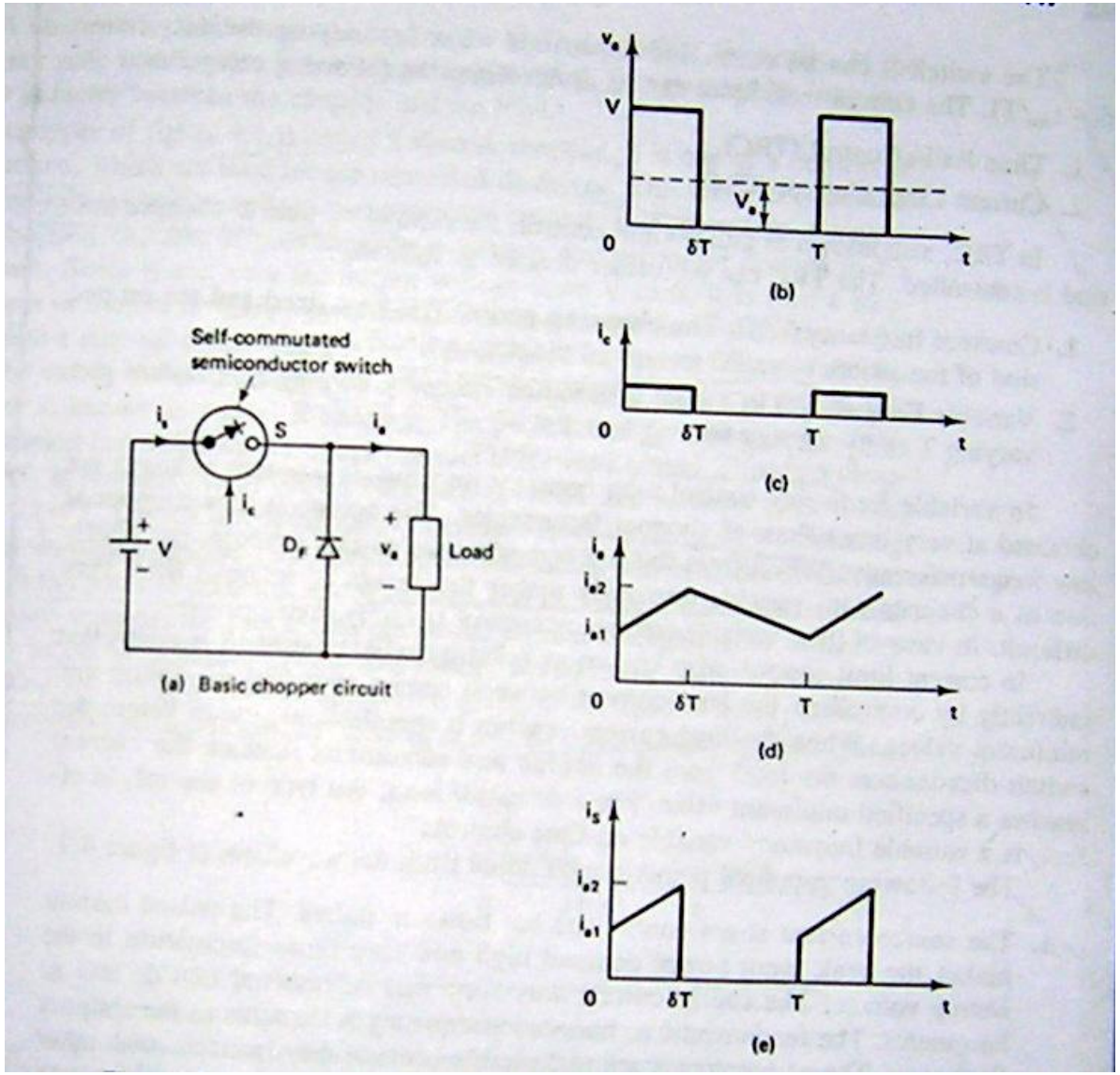
Principle of operation and control techniques.

Semiconductor switch (Thyristors/GTO/Power transistor) is switched on for a given period.

$T_{ON} = \delta T$.

δ =Duty Cycle/Duty Ratio.

CLASS A -STEP DOWN CHOPPER



Current increases from I_{a1} to I_{a2} when applied voltage is V.

From δT to T current decreases.

$0 \leq t \leq \delta T$ Duty interval

$\delta T \leq t \leq T$ Freewheeling interval

$V_{dc} = \delta V$.

The control techniques can be divided into two catagories

- Time RatioControl(PWM)
- Current Limit Control

In TRC the ratio of ON period to chopping period is controlled.
It can be further divided into

- ✓constant frequency TRC
- ✓Variable Frequency TRC

One-quadrant operation of chopper fed dc drive

Step down chopper (Class A)

Class A chopper can be utilized for speed control of dc motors.

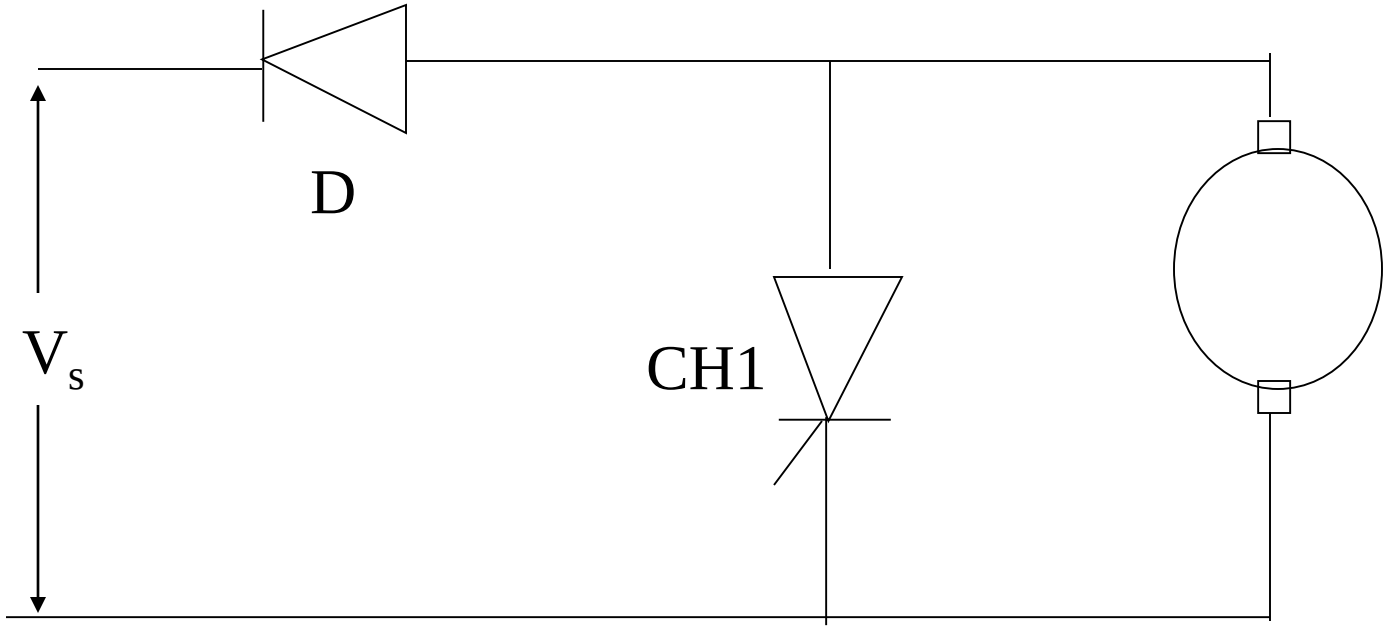
During chopping a step down action take place and hence the voltage can be controlled.

Applied voltage is +ve, hence current drawn is +ve.

Torque developed is +ve and motor rotates in forward direction.

Hence single quadrant operation (quadrant I).

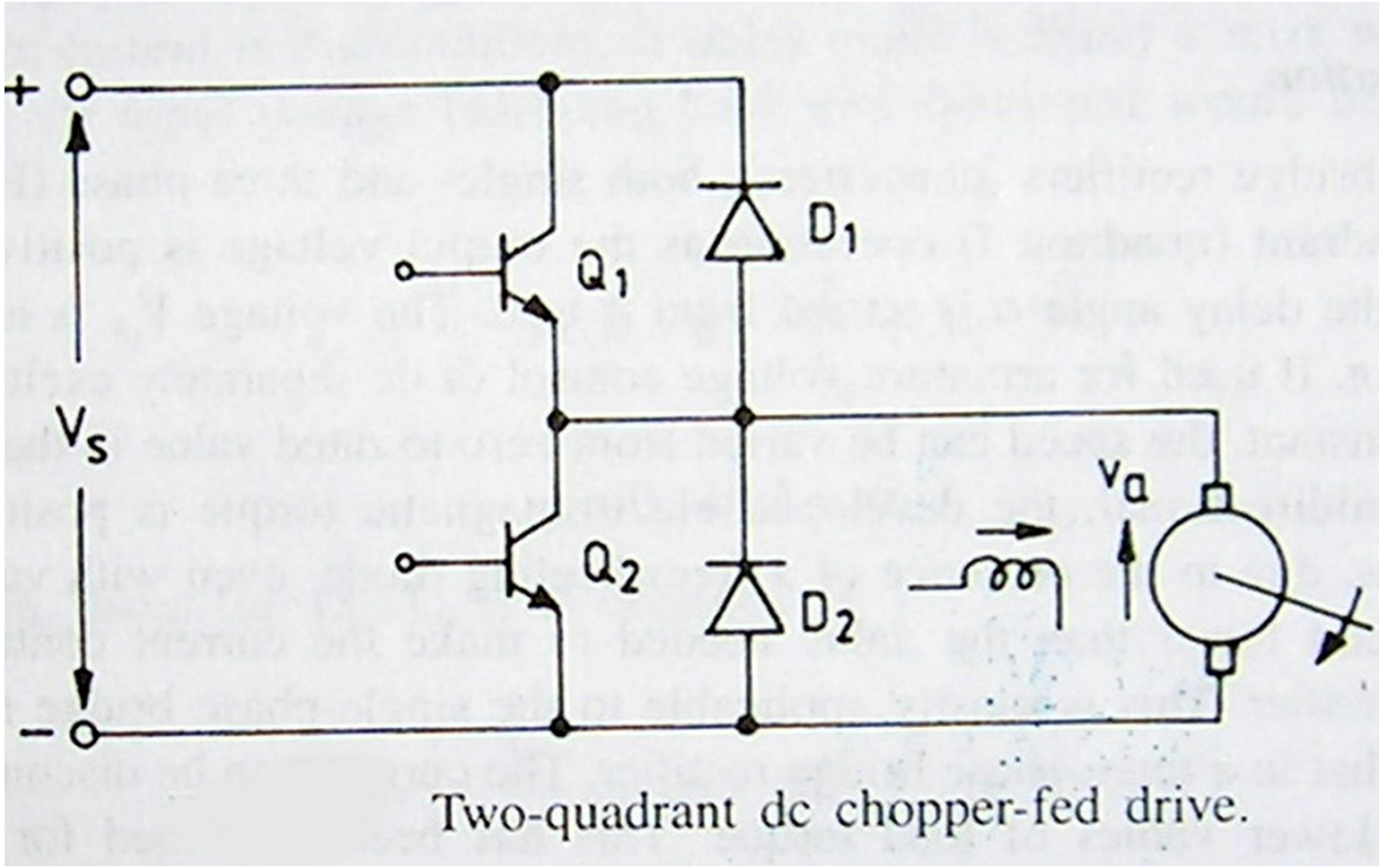
Step –up chopper (class B)



- The positions of switch and diode is interchanged compared to Step down chopper.
- As motor run as generator,CH1is switched On.
- Hence current build up in armature inductance.
- When CH1 switched OFF,current tends to flow in the same direction due to effect of inductance.
- i.e. current fed back to source.
- Motor slows down due to this regenerative braking.

- Step up and step down choppers gives one quadrant operation
- We can combine the se to give Two/Four quadrant operations.

Two Quadrant Chopper-fed drives



Two-quadrant dc chopper-fed drive.

Q1 and diode D2 can be utilized to have operation of the motor in first quadrant.

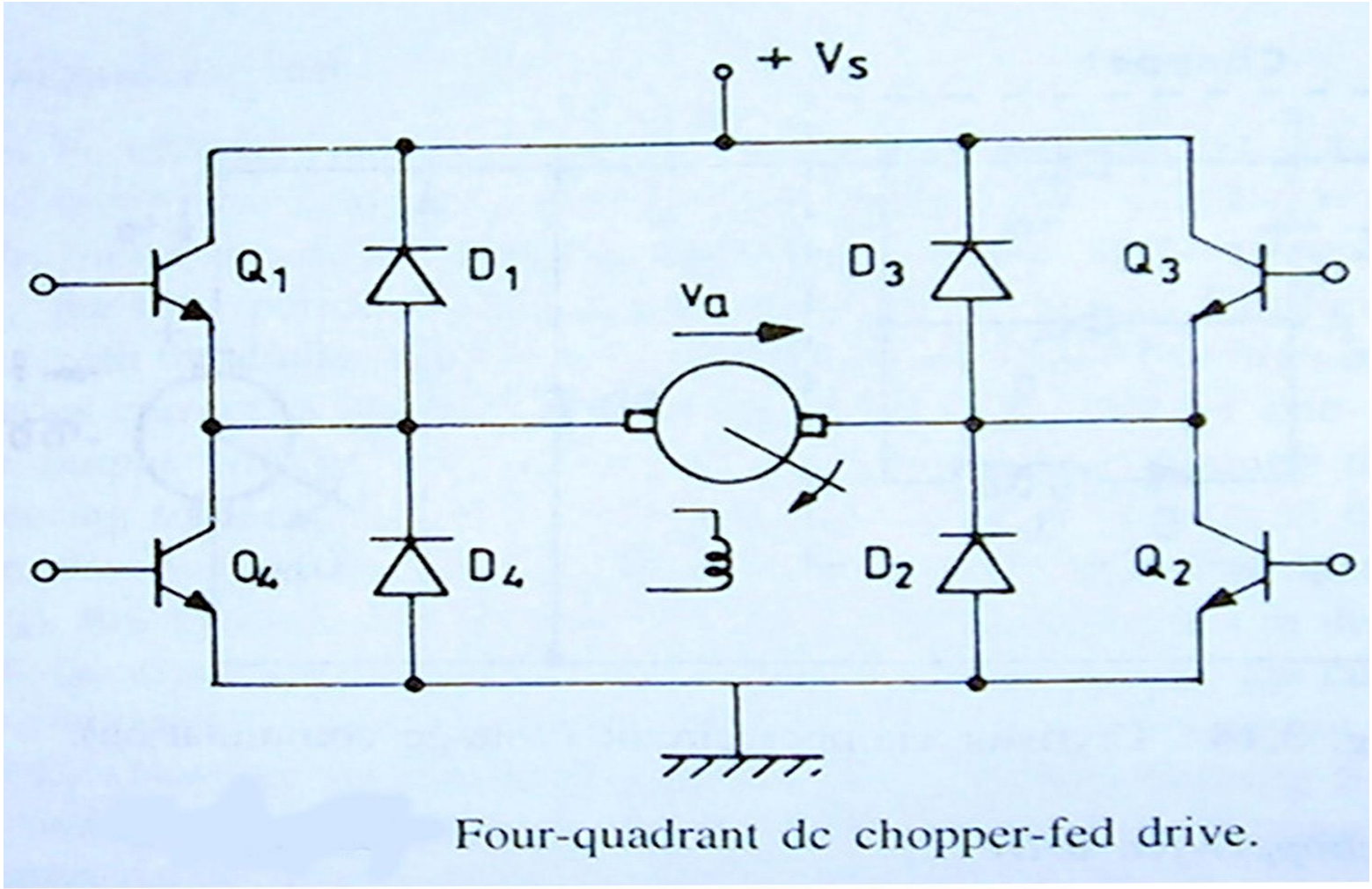
If Q1 is switched on motor will be in forward motoring mode.

When Q1 is switched off D2 will act as a freewheeling diode.

Similarly if Q2 and D1 is operated ,the chopper will act as a step up chopper.

Hence the machine will work in the regenerative braking mode.

Four Quadrant Chopper Fed Drives



Four-quadrant dc chopper-fed drive.

Quadrant –I (forward motoring)

Q1 and Q2 are turned on together ,it is ON time

Q1 or Q2 ,or both Q1 and Q2 can be switched off.

If Q1 is switched off current flows throughQ2,D4.

If Q2 switched off current flows throughQ1,D3.

If both switched off current flows through D3,D4

Quadrant-II (forward braking)

Q1,Q2,Q3 are OFF Q4 ON.

Current flows through Q4 ,D2.

When Q4 if turned off current flows through diode D! and D2.

Hence returning energy to source.

Quadrant –III (Reverse Motoring)

Q3,Q4 are turned on and then turned off.
Current flows through D1,D2

Quadrant-IV (Reverse Braking)

Q2 turned ON, Other Transistors OFF.